WINDOW REPORT: Claude (Claude Code, model `claude-opus-5-5`) | "3D glove model for Misanthropic Cow" | 2026-10-28

Window code: GLOVES · **Session:** `session_01TFV9tjKBmiqmMVJECeYn3N` (claude.ai cloud session, origin `web_claude_ai`) · **Repository:** `eccentricexhibits/main`, branch `claude/misanthropic-cow-glove-model-i5406l`, folder `Misanthropic_Cow_Gloves/`

> **Visibility notice (read first).** This conversation exceeded the model's working context once. At **2026-10-28 23:33:32 UTC** the earlier part (18:22–23:33 UTC) was replaced in my working memory by an automatic summary. However, the complete session transcript (JSON lines, 2,996 events, 63.2 MB) is still stored in the container, together with every uploaded file, every script and the git history. **This report was reconstructed from those primary records**, not from memory. Consequences: - Sections 1–7 and 11 for the period before 23:33 UTC are drawn from the transcript, tool outputs, files and commits, and are marked [RECORDED] where read directly from them. - My internal reasoning ("thinking") for that period is only partly stored (30.7 k characters survive). Rationales for that period are therefore taken from what I *wrote* to the user at the time, and are marked as such. - Timestamps are **UTC** as recorded by the system. **Local time is not visible** [UNKNOWN time zone]. - The report covers this window only. The current exchange (the documentation request) was still running when the token figures were read.

Labels: **[RECORDED]** visible in the conversation, tool output, files or git; **[INFERRED]** my deduction, with the reason; **[UNKNOWN]** not determinable; **[ESTIMATED]** a numeric estimate with its method stated.

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0. Window summary

Item	Content
AI program and model	Claude Code (Anthropic) running in a claude.ai cloud container; model **`claude-opus-5-5`** (Opus 5.5) for all 985 assistant messages [RECORDED: transcript `model` field; session `configured_model` and `last_served_model`]. Effort level "high", permission mode "auto" [RECORDED: session metadata]. Claude Code container version 2.1.316 [RECORDED].
Other AI used in this window's pipeline	**ChatGPT** image generation (run by the user from prompt kits written here): three glove views and two texture tiles [RECORDED]. ChatGPT model/version [UNKNOWN]. **Tripo** image-to-3D multiview (run by the user): one model [RECORDED]; Tripo version and settings [UNKNOWN]. **Gemini**: not used in this window [RECORDED: no mention; the kit README says the prompts "work in ChatGPT image generation or Gemini", but the user used ChatGPT].
Purpose / pipeline stage	Build the character's **costume gloves** (left + mirrored right): references → ChatGPT multiview → Tripo → Blender cleanup, split into animatable finger pairs, optimisation, UVs, bakes, "slipper look" materials from AI tiles, terry smoothing, optional true displacement, and integration notes for the full-character window.
Received from earlier work	`3D_Model_Pipeline_PLAYBOOK.zip` (19.2 MB) and `PLAYBOOK.md` (20 KB) from the earlier **slippers** window (process, scripts, worked example); 16 reference photos (2 RAR parts); later the mask scan `cowMask1.stl` (2014 MakerBot Digitizer scan per the user's master-document brief; the file itself carries no date) [RECORDED: uploads].
Handed off	Final glove pair (v4, 20 mm terry smoothing) as GLB + baked .blend + live-material .blend + displacement .blend, renders, all scripts, `HANDOFF_gloves.md`, `PROJECT_STATUS.md`, `INTEGRATION_blender_render.md`; five handoff zips each < 29 MB; everything committed and pushed [RECORDED].
Time range	First message **2026-10-28 18:22:55 UTC**; last production work **2026-10-29 01:21:00 UTC** (integration note sent); this documentation request **2026-11-01 01:35:49 UTC** [RECORDED]. Session created 2026-10-28 18:22:45 UTC [RECORDED].

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1. Prompt log

Token columns are **[RECORDED]** from the per-request `usage` fields in the transcript (364 API requests in total, de-duplicated by request ID, taking the final streamed values). "Output" = tokens the model generated (visible text + thinking + tool-call code). "Context processed" = input + cache-read + cache-write tokens summed over every request in the exchange; it is large because each request re-reads the whole conversation (mostly from cache).

#	Timestamp (UTC; local not visible)	User prompt (verbatim excerpt + summary)	What the AI did	Outputs (figure IDs)	Tokens for this exchange
1	2026-10-28 18:22:55 (+ intake answers 18:26:47; stop-hook notice 19:09:04)	*"Read PLAYBOOK.md and use this process to build a 3D model of gloves for my character "The Misanthropic Cow". I've supplied reference photos and handoff files from a previous Claude window..."* + 4 attachments (playbook zip, PLAYBOOK.md, 2 RAR parts). Intake (dialog): "No, use AI views"; "Worn, hoof closed, but please keep both hooves separate (I would like to be able to animate both parts separately)"; "Satin black, grey wear"; "Static, one mesh".	Unpacked archives (installed `unar`); viewed all 16 photos; asked 4 intake questions; installed the 3D toolchain; built white-background cut-outs with rembg (isnet failed → birefnet), inpainted a ribbon, made 5 detail crops, a 10-point design key, a 5-step ChatGPT prompt kit + fix-ups, a guide page (published artifact); committed `8a284f0`. Stop hook flagged an untracked stray wheel file; the AI deleted it.	F030–F060 (working crops, segmentation tests, cut-outs, design key, guide page screenshot); kit zips	59 requests; output **70,350**; context processed 10,933,759
2	19:11:46	*"I've got a 3D scan of the mask, and I've got two pictures of the gloves next to the mask. Does that help for scale?"* + `cowMask1.stl` + 2 photos.	Rendered/analysed the STL (installed EGL libs), sphere-fit the bead eyes, matched photo 2 to the EXIF-tagged D800 frame, computed camera height and real glove size, rebuilt the kit with measured proportions, updated the guide page; commit `7360511`.	F017, F018 (inputs); F061–F076	30 requests; output **45,327**; processed 9,590,949
3	20:24:30	*"Here's the first image that came back; the hooves are attached together. Is this correct? I was looking to animate both parts separately. Is this a problem with our prompts or ChatGPT?"* + 1 image.	Explained that the closed hoof is intended (split happens in 3D), checked the view against the checklist (hoof 36 % of height, ~3.1× tall/wide), approved it.	F019	2; out 2,844; processed 721,431
4	20:27:23	*"Is there a reason why we're skipping step 2, and keeping the same window open? ..."*	Explained the step numbering (Step 1 makes no image; the front view *is* Step 2) and why one ChatGPT chat is used.	–	1; out 1,350; processed 363,684
5	20:35:13	*"How are these?"* + 3 images (front re-sent, little-finger side, thumb side).	Checked consistency (hoof 35–36 % in all; side silhouettes IoU 0.80), mirrored the front to make the palm view, normalised all four to 1536 px squares, found and fixed a Left/Right slot error in the kit, sent the Tripo zip; commit `e30970b`.	F020–F022 (inputs); F077–F082	8; out 11,864; processed 3,057,170
6	20:46:38	*"here's the result!"* + 2 RAR parts (Tripo download).	Extracted (the 313-byte `.mtl` failed, recreated), measured the 1.86 M-triangle OBJ, rendered turnarounds, silhouette-compared against the inputs, listed defects and two decisions (cuff, thickness); commit `53b610a`.	F026–F029 (inputs); F083–F086	16; out 17,953; processed 6,471,220
7	20:59:34	*"Let's keep it closed and keep the current thickness, if you recommend it (the glove is long gone) ..."*	Welded the mesh in Blender, flattened invented seam beads, set metallic 0, fitted base and cleft planes, split the hoof into two toes (round 1: hooves only, pivot at hoof base), exported a low-res test GLB; commit `69828a8`.	F087–F094	23; out 48,882; processed 10,559,167
8	21:55:42	*"The pivot needs to go lower. The actual costume was just white gloves with painted carboard hooves that covered the index / middle fingers together, and the ring / pinky fingers together. The split must happen as if you were splitting your middle / ring fingers apart."* + 1 photo.	Rebuilt the split (round 2) as two finger pairs parting at the middle/ring web, pivot 13.9 cm above the cuff; fixed a protruding palm stub and a thumb cut; commit `c44b026`.	F023 (input); F095–F097	7; out 21,968; processed 3,594,675
9	22:03:33	*"Looks good; let's do the optimizing"*	Decimated (tested 150k/100k/64k; chose 100k per glove), unwrapped (4 UV rounds; fixed sliver-island specks), baked 4K colour/normal, painted the cut walls, exported the real-size v1 pair; offered "keep" vs "slipper look"; commit `87629ba`.	F098–F118	30; out 55,192; processed 17,542,278
10	22:26:31	*"Let's go with slipper look. Can we ensure that the hooves keep the long paint brush strokes that the actual hooves had?"*	Made rotated photo swatches (strokes vertical), wrote two tile prompts, built the geometric paint/terry region map; commit `3c6c888`.	F119–F132	11; out 19,478; processed 7,237,844
11	22:43:32	*"Are these good?"* + 2 tiles.	Prepared tiles (seams, height, roughness), built the live material, baked AO, fixed charcoal paint (gamma 1.3) and AO greying (normalise, ×0.5), final 4K bakes, v2 export, lean live .blend, renders, handoff doc, 4 zips; commit `76ca285`.	F024, F025 (inputs); F133–F154	20; out 42,335; processed 14,587,820

#	Timestamp (UTC; local not visible)	User prompt (verbatim excerpt + summary)	What the AI did	Outputs (figure IDs)	Tokens for this exchange
12	23:25:00 (context compacted 23:33:32; same message re-delivered by the user mid-turn 23:49:05)	*"The glove looks a tad too lump at the moment. Can you smooth it out?"*	Smoothing round 1–2 on the split parts (5/10 mm, pits survived), round 3 on the unsplit mesh (10 mm), full re-run of split→bake→export, before/after renders; worker restart during a bake (exit 137) recovered by baking passes separately; commit `c7cf833`.	F155–F171	59; out 54,719; processed 12,936,168
13	2026-10-29 00:04:37	*"is there some way of building the little bumpy bits that are naturally a part of the material ... or is that traditionally handled via a different process?"*	Answered: normal/bump maps (current), displacement, hair/curve loops; recommended options.	–	1; out 1,777; processed 212,477
14	00:06:45	*"The original Tripo conversion added a bunch of artifacts ... Would your new version just reuse their mesh data, or actually come up with something new?"*	Explained what is Tripo's (shape, folds, hoof grooves) and what is new (fine terry from the tile); offered stronger smoothing or a rebuild.	–	1; out 1,811; processed 214,341
15	00:07:50	*"yes, show me the stronger smoothing side by side"*	Ran 20 mm and 30 mm unsplit smoothing, split both, clay renders vs 10 mm, reported widths and side effects.	F172	7; out 3,972; processed 1,544,430
16	00:18:31	*"Let's go with 20mm."*	Full pipeline re-run at 20 mm; diagnosed white specks on the hoof (tile lint flecks + bake-margin overwrite between parts), fixed both (`bake_atlas.py`, fleck inpainting), v4 delivery, zips; commit `4e19ad4`.	F173–F194	48; out 43,645; processed 12,887,576
17	00:50:19	*"try the displacement option for the terry bumps"*	Built a displacement mask and a Cycles displacement .blend; tested 4 settings (A bump, B raw 1.5 mm, C softened 0.3 mm, D softened 0.12 mm, delivered); one out-of-memory run (1 px screen dicing); five handoff zips; commit `b520515`.	F195–F202	27; out 35,113; processed 9,334,027
18	01:14:35	*"I think this looks great, but will this only work in Blender? I plan on adding the model to a 3D model of the Misanthropic Cow that I'm doing in another Claude window."*	Explained portability (GLB anywhere; displacement Cycles-only) and integration rules.	–	1; out 1,670; processed 375,502
19	01:17:04	*"Yes, write the integration note for a Blender render"*	Tested appending into a fresh character file with a stand-in rig, verified spread signs, relative paths after moving the folder and a Cycles render, wrote `INTEGRATION_blender_render.md`; commit `9ce16ec`.	F203	10; out 20,161; processed 3,915,820
20	2026-11-01 01:35:49	*"DOCUMENTATION REQUEST: Misanthropic Cow 3D model, process report for this window ..."*	This report, the figure register and package.	report.md / report.pdf / figures	In progress when measured (≥ 3 requests; output ≥ 5,319) [RECORDED partial]

2. Inputs and source data

Input	Type / origin	Size	Resolution / polycount / units	Defects or notes
`3D_Model_Pipeline_PLAYBOOK.zip`	Earlier Claude window (slippers): `PLAYBOOK.md`, 29 scripts, templates, worked example (`HANDOFF_slippers.md`, slipper reference images incl. `real_fabric_swatches/`) [RECORDED]	19,190,005 B	–	The slipper reference images in it were not opened in this window [RECORDED: no reads].
`PLAYBOOK.md`	Same (also attached separately)	20,054 B	–	–
`Glove_Reference_Materials.part1/2.rar`	User's photo set	30,408,704 B + 16,777,083 B	16 JPEGs (table below)	Needed `unar` 1.10.1 to extract [RECORDED].

Input	Type / origin	Size	Resolution / polycount / units	Defects or notes
13 × "Original Costume - Glove Reference (n of 13).jpg" (F002–F014)	Camera photos; EXIF Model "Leaf Aptus 22"; DateTimeOriginal 2012-11-30 12:00 – 2012-12-01 16:52; processed in Lightroom Classic 15.6 [RECORDED EXIF]	2.5–5.6 MB each	4000×5336 or 5336×4000	Coloured studio light (green/pink/teal) shifts terry colour; costume shots, gloves small in frame.
"Left Hand Glove - No Hand Inside.jpg" (F001)	Camera photo; NIKON D800, 24 mm (35 mm-equiv 24); DateTimeOriginal 2013-10-31 21:16; edited in Photoshop 27.11 [RECORDED EXIF]	1,254,074 B	3159×3072	Toes parted; grey dirt mark on the terry. Key frame for front view, scale and swatches.
"Right Hand Glove - No Hand Inside.jpg" (F016)	Photo, no camera EXIF [RECORDED]	2,648,843 B	2922×2850	Did not segment cleanly; not used for cut-outs [RECORDED].
"Right Hand Glove - Alt Image - No Hand Inside.jpg" (F015)	Canon EOS 6D, EF40mm f/2.8 STM; DateTimeOriginal 2016-05-24 [RECORDED EXIF]	2,153,587 B	3465×2285	Glossy green-black hoof and a cyan drawn ellipse; excluded [RECORDED].
`cowMask1.stl`	3D scan of the mask (user: "a 3D scan of the mask"; 2014 MakerBot Digitizer per the master-document brief)	15,734,284 B	Extent ≈ 227 × 201 × 158 units, interpreted as **mm** [RECORDED tool output; unit INFERRED from size]	A relief on a rectangular base block rather than a full head [RECORDED]. Triangle count [UNKNOWN: not recorded in visible output].
Photos 1 and 2 of the gloves beside the mask (F017, F018)	Camera photos uploaded inline	272,680 B; 364,389 B	2000×1335 each (as received)	Photo 1: segmentation lost both hooves (darker); photo 2 = same D800 24 mm frame as F001 (template match 0.987) [RECORDED].
ChatGPT front view (F019; re-sent identically as F020)	AI image (ChatGPT)	146,696 B (WebP)	1172×1832	Not square (prompt asked 1:1) → padded during normalisation. Slightly glossier/marbled hoof than the real satin [RECORDED assessment].
ChatGPT side views (F021, F022)	AI images	66,850 B; 64,070 B	1254×1254	Little-finger side has diagonal "twisted sock" wrinkles [RECORDED].
User's hand photo (F023)	Phone/camera photo	261,724 B	1506×2000	Shows the Vulcan-salute split location.
`furry_glove_3d_model.part1/2.rar`	Tripo output (OBJ + PBR maps)	30,408,704 B + 29,743,216 B	OBJ 185,803,706 B: **1,863,202 triangles, 948,007 vertices** (931,603 after welding), extent X 0.482 × Y 0.972 × Z 0.258 model units (1 unit = 25.4 cm at the measured length); maps: base colour 8192², normal 4096², metallic 4096², roughness 4096² (F026–F029) [RECORDED]	`.mtl` (313 B) failed to extract; recreated [RECORDED]. Metallic hoof 0.923 mean; roughness lost on import; invented seam beads; closed flat cuff; fused toes [RECORDED].
ChatGPT tiles (F024, F025)	AI images	407,614 B; 488,262 B (WebP)	Terry 2000×2000 (the chat preview copy was 1932×1932); paint 1254×1254	Terry already seamless (edge/interior ratio < 1); paint seam in the vertical direction (ratio 1.5) and ~290 small cream lint flecks (found later) [RECORDED].

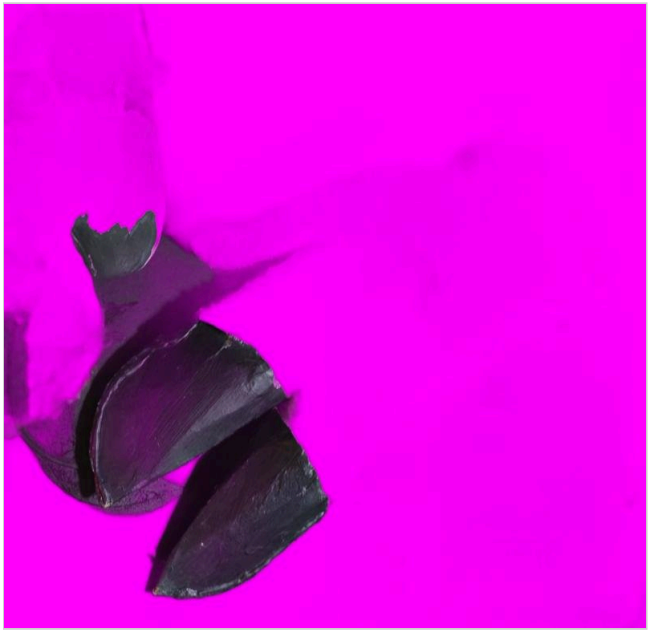
3. Methods

Environment [RECORDED]. Ubuntu 24.04.5 LTS container (claude.ai cloud), CPU only, 16 GB RAM (stated in the handoff), Python 3.13.17. Blender as a Python module: **bpy 5.2.3 LTS** (headless; scripts run with plain `python3`). numpy 2.5.3, scipy 1.18.1, OpenCV 5.0.0, Pillow 12.3.0, trimesh 5.2.0, pyrender 0.1.45 (installed `--no-deps`, EGL offscreen),

rembg 2.0.86 with onnxruntime 1.29.0 (model `birefnet-general`), networkx 3.7, rtree 1.4.1, imageio 2.38.1, `unar` 1.10.1, Chromium (Playwright-bundled) for page screenshots and this PDF. Versions were read at report time; the same container environment was used throughout [INFERRED: the main install ran at 18:42–18:44 UTC, EGL libraries were added at 19:12 UTC, and nothing was reinstalled afterwards]. Non-Blender scripts that read downloaded data were run with `python3 -I`.

Coordinate and unit conventions [RECORDED]. Work files in the Tripo model's units: glove 0.972 units tall = 24.7 cm, so 1 mm = 1/254 unit (≈ 3.93 units/m). Blender frame: Z up, back of the hand toward −Y, thumb (left glove) toward +X. Deliverables in **metres**, real size; left glove cuff centre at x = −0.12 m, right glove = mirrored node at +0.12 m. glTF: Y up, back of the hand toward +Z. Tripo slot convention: Left = the object's own left (= thumb side of this left glove).

1. **Intake (18:22–18:27).** Read PLAYBOOK.md; one 4-question intake dialog; answers in §1 row 1 [RECORDED].
2. **Reference preparation (`prep_refs_gloves.py`).** rembg `isnet-general-use` kept only the black hooves (F039, F040), so **birefnet-general** was used (F041). One cut-out job per process: several birefnet jobs in one process exhausted 16 GB [RECORDED]. Colour neutralisation strength 0.8; keep components > 15 % of the largest. A lilac ribbon in Original Costume 10 of 13 was **inpainted before segmentation**. Removing it afterwards cut a false notch (F044 → F045). Outputs: REF_A/B/C cut-outs 1536 px (F052–F054), DETAIL_1–5 crops 1024 px (F055–F059), reference sheet F049.



GLOVES-F039 · Background-removal test on the flat-lay (magenta = removed): the model kept only the black hooves and dropped the cream terry.



GLOVES-F041 · Segmentation with rembg birefnet-general: whole glove (terry + hooves) retained on magenta.



GLOVES-F044 · REF_B and REF_C cut-outs; removing the lilac ribbon after segmentation cut a false notch into the top hoof toe.



GLOVES-F045 · Cut-out after inpainting the ribbon before segmentation: clean hoof outline.

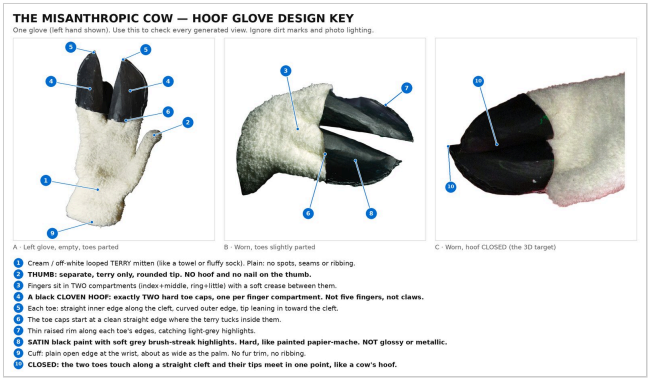


GLOVES-F049 · The three final reference cut-outs side by side: A flat left glove, B worn toes slightly parted, C worn hoof closed.



GLOVES-F052 · REF_A: white-background cut-out of the left flat-lay (1536 px).

3. **Design key (`design_key_gloves.py`).** Three panels with 10 numbered feature badges placed via grid overlays (F046–F048); output F051 (2860×1650).



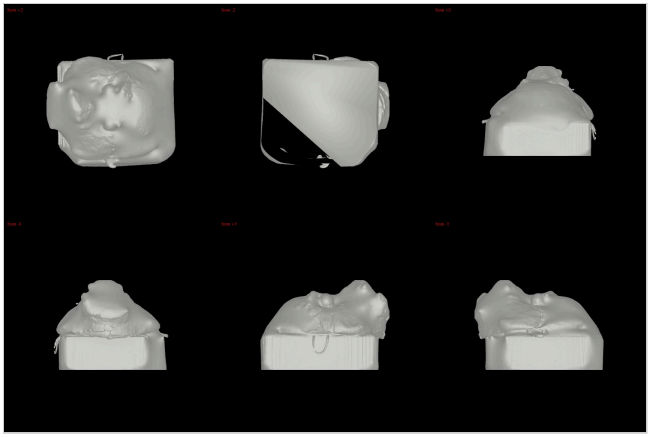
GLOVES-F051 · Annotated design key, 2860x1650: three reference cut-outs with 10 numbered features (thumb has no hoof, exactly two toe caps, satin paint, closed cleft, etc.). Sent 19:08 and attached in kit steps 1 and fix-ups.

4. **ChatGPT prompt kit (`kit_content.py` → `build_kit.py` → `build_page.py`).** Steps 1–5 + fix-ups A–J, packaged as zips with images numbered in attach order; guide page published as a private artifact "Hoof Glove View Kit" (https://claude.ai/artifact/5wPgSKLfAkQaDFmgiLiA6y; screenshot F060). Rebuilt at 19:27 (measured proportions) and at 20:36 (Tripo slot fix only). **The user ran the 19:27 version**: generation happened between 19:27 and 20:24, and the Step 1–4 prompt text is identical in the 19:27 and 20:36 builds (git diff of `kit_content.py` shows only the slot table changed) [RECORDED + INFERRED timing]. Verbatim prompts: Appendix A.

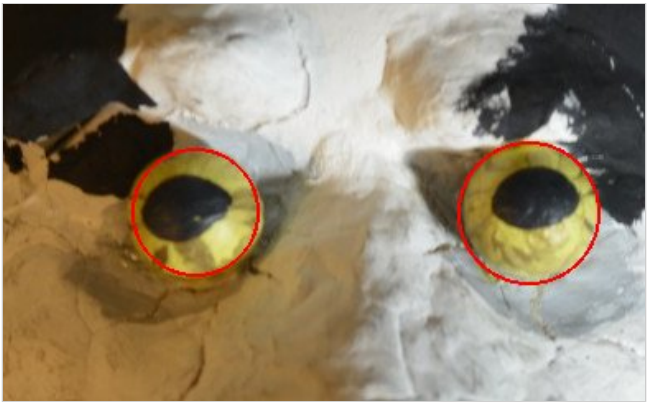


GLOVES-F060 · Screenshot of the published 'Hoof Glove View Kit' guide page (artifact) used to check layout before publishing.

5. **Real-world scale (`measure_scale.py`).** The STL is a relief on a base block (F061). A sphere was fitted to each bead eye on the scan: centre-to-centre **56.9 mm**, beads ~23 mm across. Photo 2 matched the D800 frame (template match 0.987), and its EXIF lens gives f = 24 mm on a 35.9 mm sensor, so f_px = 24/35.9 × image width. Hough circles on the yellow beads gave their pixel spacing (F066–F069), hence the camera distance to the eye plane. The eye centres sit ~65 mm above the mat (scan centre Z 122.7, rim at Z ≈ 58), so the camera was ~44 cm above the mat. Scale was evaluated at each part's height (terry ~10 mm, hoof outline ~15 mm). Landmarks were picked on gridded zooms (F072, F073); widths come from birefnet glove masks (F070). Cross-checks: beads 22.7 mm vs 23 mm; mat 598 mm vs a 610 mm (24 in) cutting mat. Uncertainty ±5 % [RECORDED]. A plotting bug drew the eye line without the crop offset (F074); the numbers were unaffected and the drawing was fixed (F075, F076).



GLOVES-F061 · Four renders of cowMask1.stl: the mask is a relief on a rectangular base block (print-base-like), not a full head.



GLOVES-F067 · Circle fits to the two bead eyes in photo 2.



GLOVES-F070 · birefnet glove masks on photos 1 and 2 (left glove red, right glove blue); photo 1 lost both hooves.



GLOVES-F076 · Measurement annotation on photo 2: bead-eye spacing, camera height ~44 cm, glove lengths 24.7 cm, terry 15.8/16.0 cm, toe caps 8.3-9.4 cm.

6. **AI views (user in ChatGPT).** The front view was the first image returned (user's words); attempts for the other views [UNKNOWN]. Checks: hoof 35–36 % of the height in all three views; side silhouettes mirror each other with IoU 0.80 [RECORDED].
7. **Tripo inputs (`normalize_glove_views.py`).** 1536×1536 white squares; glove spans y = 115…1421 px (85 % of the height); horizontally centred on the hand axis taken from the top 30 % (hoof) rows, so the thumb doesn't shift the glove. Back = mirrored front. Line-up check F077; files F078–F081 [RECORDED].

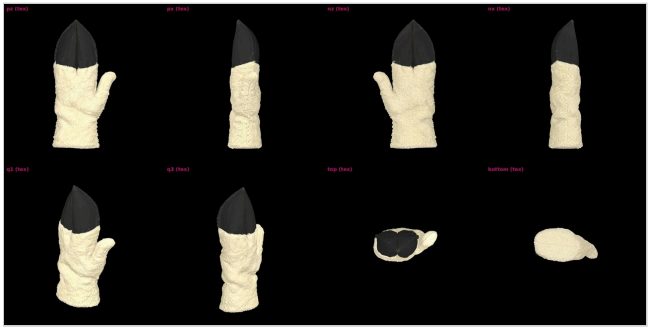


GLOVES-F077 · The four normalised Tripo inputs side by side on 1536 px squares with common top/bottom lines and centre axis.

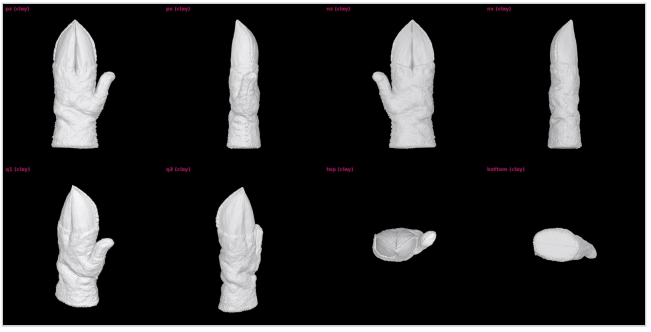


GLOVES-F078 · Tripo Front slot: approved front view normalised to 1536x1536 white square.

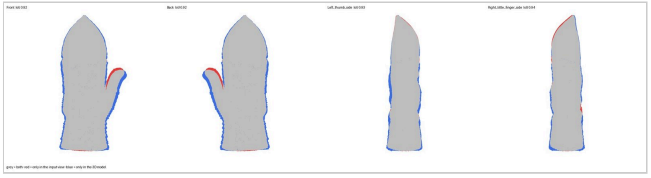
8. **Tripo evaluation (`render_model.py`, `measure_tripo_glove.py`, `compare_to_views.py`).** pyrender turnarounds (F083, F084); orthographic silhouettes vs inputs (IoU 0.92–0.94; F085); measurements in §4; close-ups of defects (F086). trimesh `split()` on 1.86 M triangles ran out of memory, so a scipy connected-components count was used instead (1 component) [RECORDED].



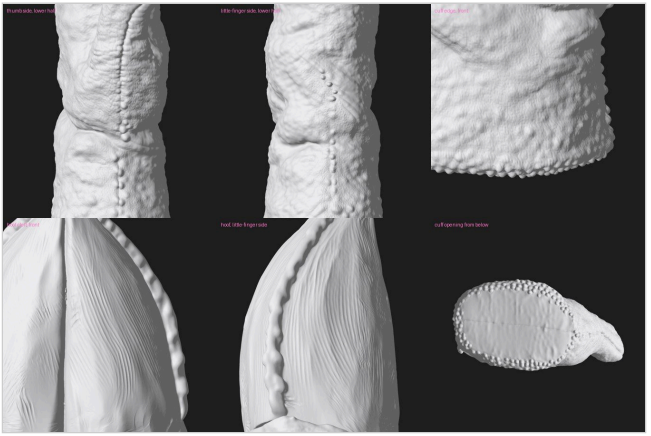
GLOVES-F083 · Tripo v1 model, textured turnaround (Tripo's own texture).



GLOVES-F084 · Tripo v1 model, clay turnaround (shape only).

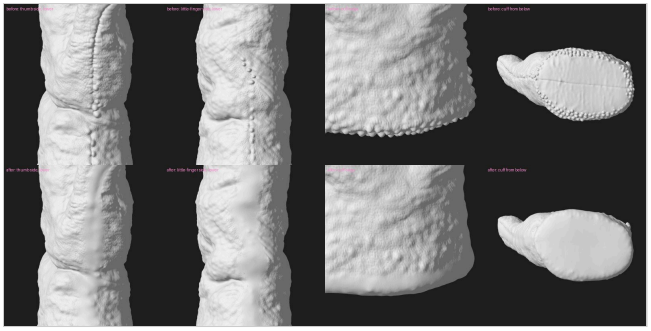


GLOVES-F085 · Tripo v1 silhouettes over the four input views (IoU 0.92-0.94).



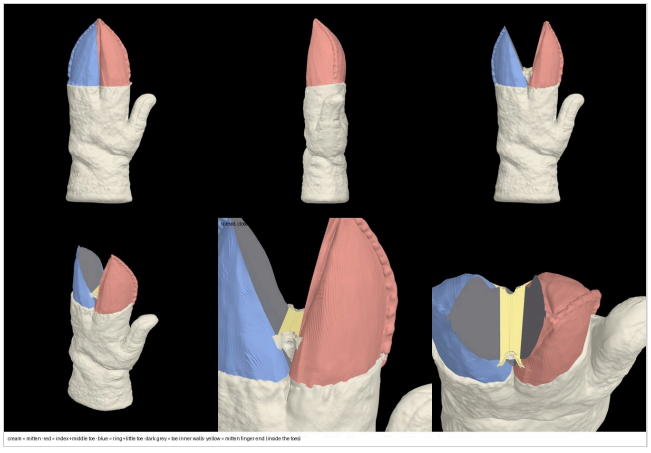
GLOVES-F086 · Tripo v1 close-ups: invented bead seams down the sides and around the cuff, closed cuff cap, hoof rim and brush grooves.

9. **Import and weld (`bl_import_weld.py`, `bl_dump_mesh.py`).** OBJ import (forward −Z, up Y), transforms applied, `remove_doubles` 1e-6 → watertight, no holes [RECORDED]. Metallic set to 0.
10. **Seam-bead flattening (`flatten_seams.py`).** Feathered band where the heavily smoothed normal faces sideways (|n_y| 0.18–0.40 smoothstep) plus the bottom 8–15 mm of the cuff, terry only (luminance ≥ 110, feathered). Vertices projected back onto a smoothed surface along the normal, 2 passes, max move 1.4 mm; the hoof's real beaded rim kept (F090) [RECORDED].



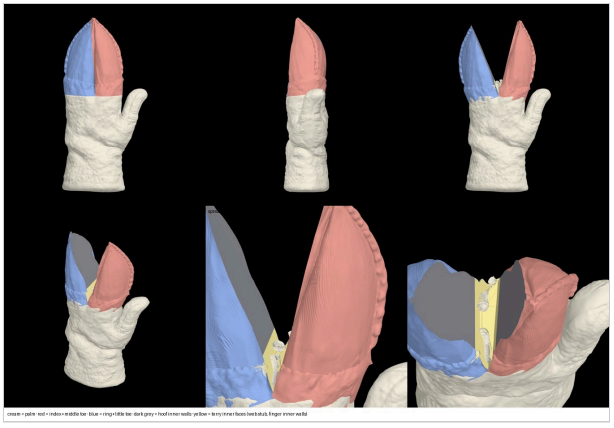
GLOVES-F090 · Bead seams before (top) and after (bottom) flattening; terry only, hoof rim kept; max move 1.4 mm.

11. **Cut planes (`fit_hoof_planes.py`).** The hoof mask is the dark vertex colour, graph-smoothed for 30 iterations, largest component kept.
    - **Base plane:** least squares through the paint edge, thumb excluded: 64.6 % of the height, 6.9° tilt, 4 mm residual. The first fit had a 34 mm residual and was fixed by the graph smoothing [RECORDED].
    - **Cleft plane:** fitted through the groove bottom found front and back at several heights; 2.9 mm residual.
    - Values: `05_cleanup_and_split/cut_planes.json` [RECORDED].
12. **Split, round 1 (superseded).** Hooves only, pivot at the hoof base, mitten extended ~2 cm inside the toes (F091–F094).



GLOVES-F093 · Split round 1 (superseded): hooves only, pivot at the hoof base; closed and spread 18 deg (colours mark parts).

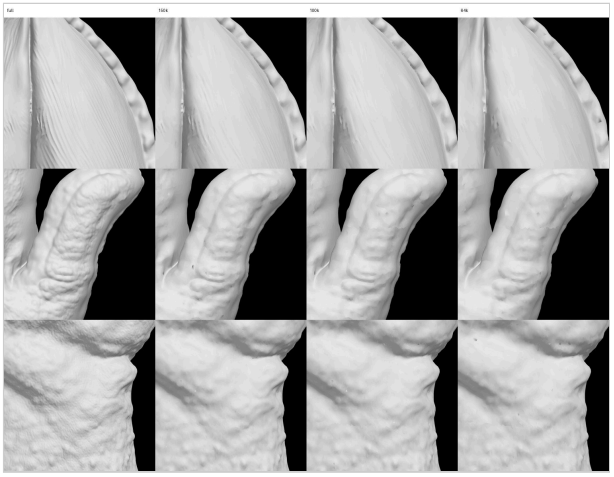
13. **Split, round 2 (`bl_split_fingers.py`, adopted).**
    - **Web plane:** parallel to the base plane, 20 mm below it, where the crease under the cleft fades.
    - **Palm:** cut flat at the web plane; the thumb is excluded from the cut (`skip_x_above` = hand x-max + 3 mm).
    - **Toes:** split by the cleft plane; openings filled flat (materials `Hoof_InnerWall` above the base, `Terry_Inner` below).
    - **Pivot:** (−0.0478, −0.0029, 0.5469) model units, i.e. 13.9 cm above the cuff, mid-thickness. `view_layer.update()` before parenting; palm stub 0 mm (a 15 mm stub poked out between the toes; F095).



GLOVES-F095 · Finger-pair split, first attempt (as viewed): palm stub poking out between the toes.

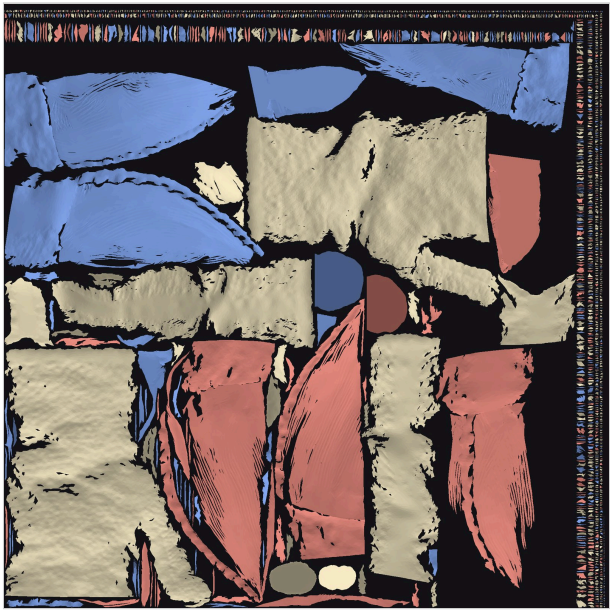
    - Thumb tip piece (21,842 faces) removed from the toe as a stray component [RECORDED].
    - Animation axes: spread = local Y, curl = local X.
14. **Optimisation (`bl_optimize_parts.py`, `compare_glove_lods.py`).** Full mesh kept as `<part>_HIGH`. LOW copy:

1. Interior vertices of hidden flat faces (cut caps, cuff bottom) decimated to 3 % of their faces.
2. One global Decimate (collapse) pass to the budget: palm 60,000 + each toe 20,000 = **100,000 triangles per glove**.
3. Compared with 150k and 64k over 8 cameras (F098) [RECORDED].

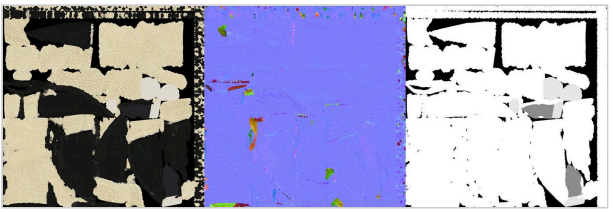


GLOVES-F098 · Decimation comparison: full mesh vs 150k / 100k / 64k triangles per glove (close-ups).

15. **UV atlas (`bl_unwrap_parts.py`).** One shared atlas for the three parts.
    - **Rounds 1–2:** smart-project at 66° and 80° (~1,800 islands; F099, F100). Sliver islands baked as specks (F101, F104).

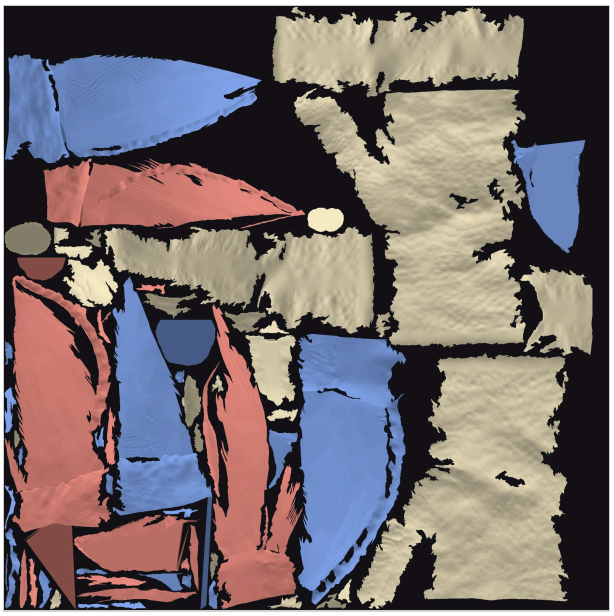


GLOVES-F099 · UV round 1: plain smart-project at 66 deg (~1,800 islands).



GLOVES-F101 · 1K test bake: black/white specks from sub-texel sliver islands.

    - **Round 3:** unwrap on Laplacian-smoothed proxies (30 iterations, λ 0.5; F103).
    - **Round 4 (adopted):** islands under 40 faces merged into their neighbour chart, seams only between charts, angle-based unwrap per chart (F105).



GLOVES-F105 · UV round 4 (adopted): islands < 40 faces merged, angle-based unwrap per chart: 64 islands, 63.9 % coverage.

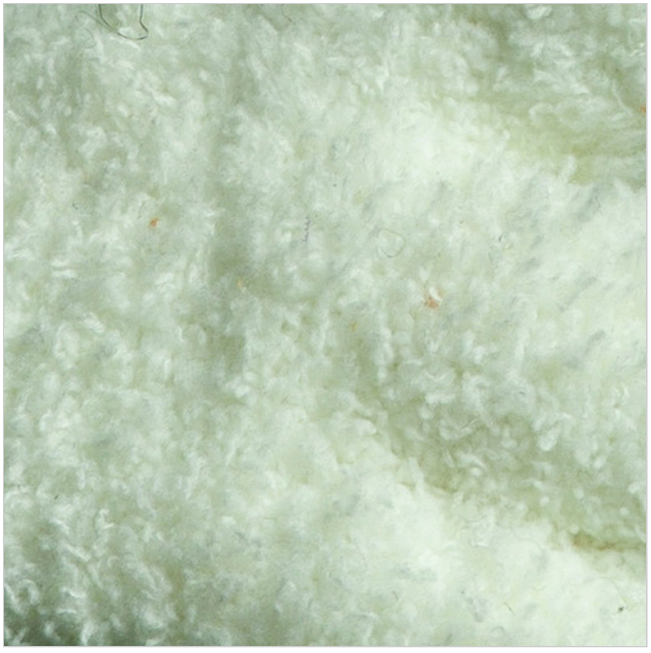
- Texel weights: hoof 1.3, terry 1.0, inner walls 0.6, cuff bottom 0.3, palm top 0.25 [RECORDED].

16. **Selected-to-active bakes (`bl_bake_parts.py`).**
    - Settings: Cycles CPU, 4096 px, 2 samples, cage extrusion 0.012, max ray 0.03, margin 16 px EXTEND; colour + tangent normal; cap edges marked sharp.
    - Cut walls painted directly into the maps (`bl_fix_cap_texels.py`; paint RGB (25,25,24), terry (223,213,179), wall roughness 0.75), because coplanar caps baked as a grey "sheet".
    - Roughness derived from colour: 0.90 − 0.35·dark, since Tripo's roughness map was lost on import [RECORDED].
17. **v1 export (`bl_export_pair.py`).** Real size 24.7 cm, metres, mirrored right glove sharing meshes; GLB 6.3 MB (F111) [RECORDED].



GLOVES-F111 · v1 pair: left glove spread 15 deg, right closed; Tripo's cleaned texture (route 9a).

18. **Tile kit (`prep_tile_refs.py`).** Swatches cropped and rotated so brush strokes run vertically (paint A rotated 50° CCW, B 125° CCW), 1024 px; prompts in Appendix A.6 (F128–F131) [RECORDED].



GLOVES-F128 · Tile kit attachment: terry loops swatch from Original Costume 10 of 13 (1024 px).



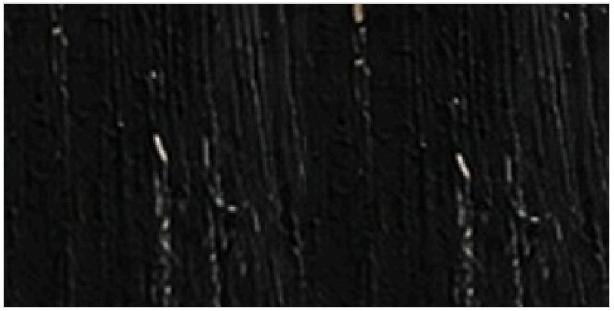
GLOVES-F130 · Tile kit attachment: brush-stroke swatch from the left flat-lay, rotated 50 deg CCW so strokes run vertically.

19. **Region mask (`bl_bake_geom_parts.py` → `make_masks_glove.py`).** EMIT bakes of world position and part ID (4096 px). Paint = finger-pair texels above the hoof base plane; Gaussian σ 0.8 px; unbaked texels take the nearest class. Region map F127 approved by the user [RECORDED: user "Are these good?" + continued].



GLOVES-F127 · Approved region map: red = painted hoof (finger-pair texels above the hoof base plane), cream = terry.

20. **Tile preparation (`prep_tiles_glove.py`).** All tiles 2048 px.
    - **Terry:** colour × (1.00, 0.99, 0.95).
    - **Paint:** vertical seam fixed by a half-offset crossfade (ramp 0.22, contrast compensation with a σ 24 blur); darks deepened with gamma 1.3.
    - **Height maps:** terry = 0.5 + 0.25·HP(σ 16); paint = 0.5 − 0.15·HP(σ 4) − 0.10·HP(σ 30), dark = raised.
    - **Paint roughness:** 0.50 + 0.22·grey-streak mask.
    - **v4 addition:** isolated bright flecks (> 100 above a 7 px median, dilated 3×3) inpainted with OpenCV Telea (radius 3, 32 px wrap padding); 288 flecks removed; seam scores unchanged (1.20/1.37 → 1.21/1.40) (F175, F176, F184) [RECORDED].

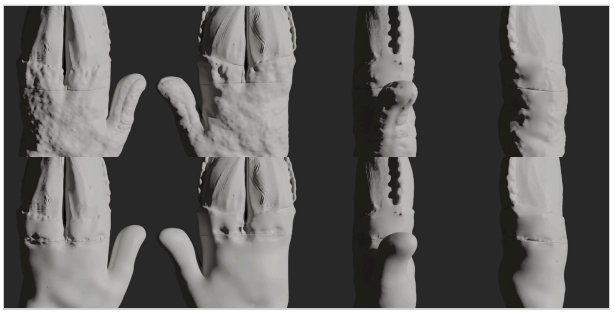


GLOVES-F175 · Pixel crops of the ChatGPT paint tile showing cream lint-like flecks.



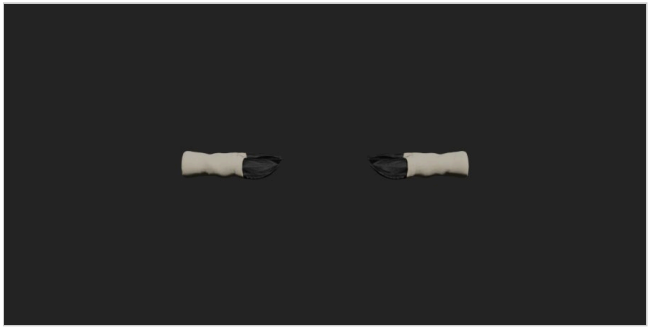
GLOVES-F184 · Prepared paint tile v4: ~290 lint flecks inpainted, strokes and grey streaks kept, 2048 px.

21. **Live material (`bl_glove_material.py`).** Principled BSDF; specular IOR level 0.4; metallic 0.
    - **Projection:** tiles box-projected from each part's **own object coordinates**, terry 4 cm and paint 6 cm, blend 0.3, so the strokes run base-to-tip and move with the toes. The UV mask switches between terry and paint.
    - **Roughness and sheen:** terry roughness 0.92 and sheen 0.45; paint roughness from its map.
    - **Bump:** tile height bump (strength 0.45, distance 0.15 cm; paint height scaled 0.45) on top of the baked detail normal.
    - **AO:** baked at 2048 px with 12 samples, distance 1.5 cm; denoised with 2 bilateral passes (9, 0.15, 6) and (9, 0.12, 6); divided by the 90th percentile; cubic upscale to 4K; gamma 0.8; multiplied in at factor 0.5. At 0.6 un-normalised it greyed the terry by ~20 % [RECORDED].
    - **Final bakes:** 4096 px, 4 samples (diffuse colour, roughness, tangent normal), ~51–58 s each.
22. **v2 export and packaging.** GLB 14.5 MB; baked .blend + `textures/`; live .blend made portable (`bl_lean_live.py`: copy textures, set absolute paths, save, make relative, save). A moved-copy test passed (F138) [RECORDED].
23. **Terry smoothing.** Rounds 1–2 smoothed the split parts (`scripts/superseded/bl_smooth_terry.py`): HIGH parts low-passed and LOW snapped with Shrinkwrap. It failed at the web (F158), because the cut walls had to stay pinned.
    - **Round 3 (adopted; `smooth_terry_unsplit.py`)** smooths the unsplit 931,603-vertex mesh **before the split**.



GLOVES-F158 · Web close-ups, original (top) vs per-part 10 mm (bottom): deep pits above the web survive.

    - Low-pass: $V' = V + w\cdot(2S(V) - S(S(V)) - V)$, where S is uniform-graph Laplacian diffusion with $(r/\bar{e})^2$ iterations ($\bar{e}$ = mean edge length, 0.31 mm). This removes lumps smaller than r while undoing shrinkage to second order.
    - Weight w = 0 on the hoof (paint side of the base plane, smoothstep 1–4 mm below it, thumb never pinned) and on the cuff bottom (feathered ~5 mm).
    - r = 10 mm (v3), then **20 mm (v4, user's choice)**, with 30 mm tested. Everything downstream was re-run (§4) [RECORDED].
24. **Speck fix (`bake_atlas.py`).** Each part is baked into a cleared image. Its own UV triangles are rasterised (OpenCV `fillPoly`, 4-bit subpixel, pixel-centre convention) to decide texel ownership. The margin is applied once with an EDT nearest fill (16 px; 4 px for geometry bakes). The ownership raster matched Blender's own part bake on 100.00 % of covered texels [RECORDED].
25. **True displacement (Cycles only).**
    - **Mask (`bl_make_displace_mask.py`):** terry only; 0 on the hoof and cut walls; smoothstep fade-in from 0.5 mm to 3 mm, by 3D distance (cKDTree) from 3,403 points sampled every 0.3 mm along the wall edges; 0.7 px blur; islands padded with EDT [RECORDED].
    - **Blend (`bl_terry_displace.py`):** Subdivision modifier type SIMPLE with adaptive subdivision in object space at a **0.3 mm** edge length (`max_subdivisions` 12). Displacement node in object space, midlevel 0.5, scale **1.2 mm**, using the terry height softened by a σ 0.12 mm Gaussian (σ ≈ 6.1 px at 2048 px per 4 cm), with half the contrast lost to the blur restored. Terry bump disabled where displaced [RECORDED].
26. **Integration test.** Appended the six objects into a fresh .blend with a stand-in armature and parented the palms to `hand.L`/`hand.R` bones. Checked spread (12° → tip gap 0.6 → 4.8 cm on both gloves), bone following (toe pivot moved 7.2 cm for a 30° bone turn), relative paths after moving the project folder (10/10 textures found), and a Cycles render (F203) [RECORDED].



GLOVES-F203 · Integration test: gloves appended into a fresh character file, parented to stand-in hand bones, rendered in Cycles.

How 2D images were related to 3D geometry.

- Photos and AI views drove Tripo's geometry.
- The photo-derived scale set real size.
- The paint/terry boundary was derived from **geometry** (a plane through Tripo's colour edge), not from image projection.
- Tiles are box-projected in object space, not photo-projected [RECORDED].

Image-generation prompts. All verbatim in Appendix A.

- **Requested settings:** square 1:1, white background, glove ~85 % of the height, near-orthographic.
- **Tiles:** square, "as large as you can make it".
- **Received:** the front view came back 1172×1832; the sides 1254×1254; tiles 2000^2 and 1254^2 [RECORDED].
- **ChatGPT model:** [UNKNOWN]. **Attempts:** front view = attempt 1 (user's words); all others [UNKNOWN].

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4. Quantitative results [RECORDED unless marked]

Real size (measured, ±5 %): cuff → hoof tip 24.7 cm (both gloves); terry cuff → hoof base 15.8 cm (L) / 16.0 cm (R); toe caps 8.5/9.0 cm (L), 9.4/8.3 cm (R); width at the wrist laid flat ~8 cm (L) / ~8.6 cm (R); thumb ≈ 4.5–5 cm; hoof ≈ 36 % of the length.

Tripo v1 (`04_tripo_v1_evaluation/measurements.txt`).

Quantity	Value
Triangles / vertices	1,863,202 / 948,007; 1 connected component
Hoof share of length	top 37 % (real 36 %)
Width incl. thumb at 5/25/45/60/70/85/95 % height	8.9 / 8.3 / 10.3 / 11.5 / 8.4 / 6.9 / 3.3 cm
Thickness at the same heights	6.5 / 5.7 / 5.8 / 5.5 / 5.3 / 4.5 / 2.3 cm
Cleft groove depth (front/back) at 70/80/90 %	9.3/7.0, 7.3/6.4, 2.6/2.8 mm; hoof thickness 53/50/36 mm
Metallic	mean 0.335, hoof mean 0.923, terry 0.133
Roughness	mean 0.840, hoof 0.750, terry 0.871
Silhouette IoU vs inputs	0.92–0.94

Split parts (v2 build).

- Palm 1.24 M faces; toes ~0.32 M faces each; toe lengths 11.1 cm (index-middle) and 10.8 cm (ring-little); pivot 13.9 cm above the cuff [RECORDED: v3/v4 runs print palm 1,238,317 / 1,237,597 faces etc.].
- Low-res test GLB: 1,794,516 B.

Decimation (8 cameras vs the 1.88 M-triangle split mesh).

Budget per glove	Mean shading diff (0–255)	Silhouette mismatch
150k	2.58	0.05 %
100k (chosen)	3.03	0.06 %
64k	3.57	0.08 % (hoof rim jagged)

UV atlas. v2: 64 islands, 63.9 % coverage; v3: 58 islands, 65.0 %; v4: 76 islands, 63.8 % (all 4096^2).

Bake timings (CPU).

- Selected-to-active, 4K, 2 samples: colour ~47–63 s, normal ~44–50 s, roughness ~53 s.
- Final material bakes, 4 samples: 51–58 s each. AO 2K, 12 samples: 24–48 s.

Terry smoothing (unsplit mesh, 931,603 vertices, mean edge 0.31 mm).

Radius	Iterations	Terry move mean / p99 / max	Width incl. thumb	Outcome
0 (v2)	–	–	12.0 cm	lumpy
10 mm (v3)	1,065	0.45 / 1.34 / 2.44 mm	12.0 cm	folds kept
20 mm (v4)	4,260	0.92 / 3.12 / 5.68 mm	11.8 cm	most folds gone
30 mm (test)	9,584	1.58 / 5.59 / 9.67 mm	11.4 cm	thumb slimmer, wobbly cuff edge

Hoof maximum move 0.000 mm in all runs. Per-part smoothing (rejected), 10 mm: palm mean 0.46 mm / max 7.55 mm; toes 0.82 / 13.72 mm, concentrated at the web.

Specks. Colour outliers on the paint (> 90 from a 5 px median): 402 → 119 after the tile fleck fix. Remaining bright patches were margin overwrite: e.g. texel (2126, 3521) went from RGB (213, 204, 175) to (21, 21, 21) after the compositing fix. Bright blobs (> 185) in the hoof close-up: 0.

Displacement. One run with 1 px screen-space dicing was killed (exit 137, out of memory). With object-space dicing at 0.3 mm: peak memory ~6.0 GB (sampled with `free -m` every 5 s); 4 close-ups at 800 px and 32 samples took 2 min 19 s. Mask covers 71.3 % of covered texels at > 0.5 (padded; the first unpadded value was 56.0 %).

Final deliverables (v4).

- `Glove_pair.glb` 15,047,220 B, 100k triangles per glove, 4K colour/normal + 2K roughness JPEG.
- Baked .blend 4.22 MB; live .blend 4.23 MB; displacement .blend 4.23 MB.
- Handoff zips: Part1 22.9 MB, Part2 14.4 MB, Part3 16.9 MB, Part4 18.5 MB, Part5 28.4 MB.

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5. Decisions, assumptions and judgment calls

Decision	Rationale (as stated at the time)	Approved by the user?
AI views instead of a photo re-shoot	Real gloves no longer available	**Yes** (intake)
Worn, hoof closed, toes separate for animation	User's specification	**Yes** (user's own answer)
Satin black hoof with grey wear	Matches studio photos, not the glossy "Alt Image"	**Yes** (intake)
"Static, one mesh" (no rig)	Intake answer; later superseded by the user's request for separable finger pairs (parented objects, still no armature)	**Yes**
Generate the hoof **closed and joined**; split in 3D	Tripo fuses touching parts; a parted hoof bakes in the pose	User asked, accepted the explanation [RECORDED]
One ChatGPT chat for all views	Consistency; slipper precedent	User asked; accepted ("It's fine if 1 window provides better results")
Palm view = mirrored front; thumb view optional	Mirrored approved view is better Tripo input	Implicitly (user followed the kit) [INFERRED]
Measure scale from photo 2 + scan (not photo 1)	Photo 1 lost hooves in segmentation; photo 2 has lens data	Not explicitly approved [RECORDED: no objection]
Accept Tripo v1 without round 2	IoU 0.92–0.94, proportions within 1 %	Implicitly (user proceeded)
Cuff closed; thickness as Tripo	Sleeve hides the cuff; no hand close-ups exist	**Yes**
Flatten seam beads (terry only; keep hoof rim)	Beads are Tripo inventions at view seams	Shown; not objected
Split as two **finger pairs** at the middle/ring web	User's correction about the real costume	**Yes** (user's instruction; "Looks good")
Web 20 mm below the hoof base	Where the modelled crease fades	**Yes** (approved split)
100k triangles per glove	Lowest budget that kept the rim beads	User approved optimising; budget not separately approved
One shared 4K atlas for 3 parts	Right glove reuses the maps	Not separately approved
Slipper look (ChatGPT tiles), keep long brush strokes	User's choice	**Yes**
Paint region from geometry (plane), not colour	The real paint edge was a clean cut	Region map shown; user proceeded [INFERRED approval]
Paint gamma 1.3; terry warmed; AO normalised × 0.5	Paint read charcoal; terry read grey; AO greyed the glove	Not separately approved
Smooth the terry (user asked), on the **unsplit** mesh	Pinned walls left pits in the split-part method	Requested; method not separately approved
20 mm smoothing	User's choice after a 10/20/30 mm comparison	**Yes**
Fix specks via per-part bake compositing and tile fleck removal	Root-caused bugs	Not asked (bug fix)
Displacement variant D (1.2 mm, softened 0.12 mm, 0.3 mm dicing) as a **separate** file	B was spiky, C patchy, 1 px dicing exhausted memory	User: "I think this looks great"
Split handoff into 5 zips	Keep each file < 29 MB (user's standing rule)	Rule from user
Assumptions: hoof side profile (slim wedge) and palm = mirror of back; thickness 5.3–5.8 cm from AI views	No photos show these	Flagged; not verified [RECORDED]

6. Problems, failed attempts and fixes

#	Problem	Diagnosis	Fix / outcome
1	RAR archives would not open (`rarfile` wheel download attempt)	No RAR tool	Installed `unar`; the stray downloaded wheel was flagged by the git stop hook (19:09:04) and deleted
2	rembg `isnet-general-use` dropped the cream terry (F039, F040)	Low contrast cream vs light background	Switched to `birefnet-general` (F041)
3	birefnet out-of-memory when several jobs shared a process	16 GB limit	One job per process
4	False notch in the hoof after removing the ribbon (F044)	Masking after segmentation	Inpaint the ribbon **before** segmentation (F045)
5	Kit size guess (30 cm) wrong	No measurement	Scale from mask scan + photo 2 → 24.7 cm; kit rebuilt (19:27)
6	Measurement image: eye line misdrawn (F074)	Crop offset omitted in drawing only	Fixed (F075)
7	Tripo Left/Right slots swapped in the kit	Convention mismatch with the slippers	Kit fixed at 20:36 before the user uploaded to Tripo
8	`.mtl` failed to extract; metallic map at a different resolution broke a lookup; thumb test caught the cuff	Archive and script issues	`.mtl` recreated; lookups fixed
9	trimesh `split()` OOM on 1.86 M triangles	Memory	scipy connected components
10	pyrender needed EGL; networkx build failed	Packaging	apt EGL libs; pyrender `--no-deps`
11	OBJ import stored a rotation as object transform	Importer behaviour	Transforms applied
12	Base-plane fit residual 34 mm	Noisy colour edge	Graph-smoothed hoof mask → 4 mm
13	Split round 1: area missing where the toes meet the mitten extension (F091)	Extension shrink direction OK; toe region missing	Rendered parts separately (F092); fixed (F093)
14	Children landed offset after parenting	Stale matrices	`view_layer.update()` before parenting
15	**Round 1 → Round 2 split** (user: pivot must be lower; caps glued to finger pairs)	Construction misunderstanding	Finger-pair split at the web (F097)
16	Palm stub poked out between the toes (F095); web plane cut the thumb	Stub folds in the crease; thumb rises above the web	Stub 0; thumb excluded from the cut
17	Mixed quads/triangles broke `foreach_get`	Bisect creates quads	Read faces through loops
18	**UV rounds 1–4:** ~1,800 islands, sliver islands baked as black/white specks (F099–F104, F112)	Sub-texel islands	Proxy smoothing + chart merging + angle-based unwrap: 64 islands (F105)
19	Coplanar cut caps baked as a grey translucent sheet; shading line at the web	Selected-to-active on coplanar caps	Walls painted into the maps; cap edges sharp
20	Tripo roughness lost on import	Importer	Roughness derived from colour
21	Paint read charcoal; AO greyed terry ~20 %; terry read grey (F135)	Tile values and un-normalised AO	Gamma 1.3; AO ÷ p90 at 0.5; terry B × 0.95
22	Handoff Part 1 (34.7 MB) and Part 4 (32.8 MB) over 29 MB	Packaging	Rebalanced (21.2/13.9/28.8/26.5 MB)
23	**Smoothing round 1 → 2 → 3:** Taubin too weak; per-part low-pass left the toe band lumpy (pins); relaxed pins still left deep pits above the web (F158)	Cut walls must stay pinned	Smooth **before** the split (F159)
24	Script crashes: stale vertex-group handle after `modifier_apply`; slicing a bpy collection	API behaviour	Look up by name; `foreach_get` arrays
25	Stale `matrix_world` (identity) in a hidden collection made a move check misleading	Hidden objects not evaluated	Unhide + `view_layer.update()`; results unaffected
26	Worker restart killed a 4K bake (exit 137)	Process killed (memory or restart) [INFERRED]	Baked passes separately
27	**v4 white specks** on the hoof (F173, F174)	Hypothesis 1: ChatGPT tile lint flecks (F175), fixed (F176) but specks persisted (F177, F178). Pixel crops (F179) + raycast to UV: a later part's 16 px bake margin overwrote an earlier part's hoof texels	Per-part bake compositing (`bake_atlas.py`) → specks gone (F180, F181)
28	Displacement 1 px screen-space dicing: out of memory (exit 137)	3 cm close-up diced to 0.04 mm/px	Object-space 0.3 mm dicing (~6 GB peak)
29	**Displacement A → B → C → D:** raw height gave spikes and dark slivers (B); 0.3 mm softening gave patchy nubs (C)	Detail finer than micro-polygons; softening ≈ loop size	0.12 mm softening (D) (F198, F202)

#	Problem	Diagnosis	Fix / outcome
30	UV seams would dip under linear filtering of the mask	Empty texels around islands	EDT padding of the mask
31	Render-time ratio claim not supported (parallel runs)	Measurement confound	Reworded with an absolute timing
32	Integration note: texture count misstated (12 vs 10)	Checked against files	Corrected

Abandoned: hooves-only split (round 1); per-part smoothing (`scripts/superseded/bl_smooth_terry.py`); 30 mm smoothing; screen-space dicing; raw/over-softened displacement; `Glove_L.glb` single-glove export dropped from the final set (identical size to the pair, since both gloves share meshes; it remains in the superseded `07_deliverables_v1/`) [RECORDED].

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7. Outputs

Deliverable	Format	Specs	Purpose
`09_final/glb/Glove_pair.glb`	glTF binary	15,047,220 B; 2 gloves (right = mirrored node sharing meshes); 100k tris/glove; 4K colour (AO baked in) + 4K tangent normal + 2K roughness; metallic 0; metres, Y-up	Any 3D software / engine
`09_final/blend_baked/Glove_pair_baked.blend` + `textures/` (3 maps)	Blender 5.2	Same as the GLB; relative paths	Blender (Eevee/Cycles)
`09_final/blend_live/Glove_pair_live.blend` + `textures/` (10 files)	Blender 5.2	Live tile material `Glove_Terry_Hoof_live` (box projection, mask, bump, AO)	Editing / renders
`09_final/blend_live/Glove_pair_live_displace.blend`	Blender 5.2 (Cycles)	Adds adaptive Simple subdivision 0.3 mm + 1.2 mm terry displacement	Close-up Cycles renders
Objects (all files)	–	`Glove_L_Palm` (root; origin = cuff bottom centre) → `Glove_L_Toe_IndexMiddle`, `Glove_L_Toe_RingLittle`; same for `Glove_R_*`; pivot 13.9 cm above the cuff; spread = local Y (Index +, Ring −), curl = local X	Animation
`09_final/renders/*`	JPEG	Final pair, close-ups, live-vs-baked, region map, before/after smoothing, bump vs displacement, pair with displacement	Review
`07_deliverables_v1/`	GLB/.blend	v1 (Tripo texture, 6.3 MB) – superseded	History
`05_cleanup_and_split/Glove_L_split_test_lowres.glb`	GLB	Low-res split test, model units	Approval of the split
`HANDOFF_gloves.md`, `PROJECT_STATUS.md`, `INTEGRATION_blender_render.md`, `01_design_brief.md`	Markdown	Process, decisions, reproduction, integration	Handoff
`02_chatgpt_kit/` (5 step zips + fix-ups + guide page), `08_fabric_tiles/` (tile kit, raw and prepared tiles, mask, region map), `03_tripo_inputs/`	Zip/PNG/JPEG/TXT	Prompt kits and AI inputs	Reproducibility
`scripts/` (45 scripts + `superseded/`)	Python	Entire pipeline	Reproducibility
Handoff zips (final set)	Zip	Part1 CONTEXT 22.9 MB · Part2 FINAL_GLB 14.4 MB · Part3 BLEND_BAKED 16.9 MB · Part4 BLENDS_LIVE 18.5 MB · Part5 SOURCES 28.4 MB (`high_full.glb`, the cleaned 20 mm-smoothed split full-detail mesh)	Transfer between chats
Artifact "Hoof Glove View Kit"	Private claude.ai page	https://claude.ai/artifact/5wPgSKLfAkQaDFmgiLiA6y	Step guide with copy buttons
13 git commits `8a284f0` … `9ce16ec`	Git	Branch `claude/misanthropic-cow-glove-model-i5406l` (no PR opened)	Version history

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8. Division of labour

Party	Contribution
User (human)	Supplied the photo set, the slipper playbook package, the mask scan and two scale photos; answered the intake; ran ChatGPT (3 views, 2 tiles) and Tripo (1 model) using the kits; approved/selected views; made the key design corrections (separate animatable toes; finger-pair split at the middle/ring web with a reference photo; keep cuff closed and thickness); chose the slipper look and long brush strokes; requested smoothing and chose 20 mm; asked for displacement and the integration note; set the 29 MB file rule.
This AI (Claude, claude-opus-5-5)	Planning against the playbook; all image processing (cut-outs, crops, design key, normalisation, swatches); scale measurement; all prompt writing; Tripo evaluation; all Blender/Python geometry work (weld, seam flattening, plane fitting, split, decimation, UVs, bakes, masks, materials, smoothing, displacement); diagnosis and fixes; renders and comparison sheets; packaging, documentation, git commits.

Party	Contribution
ChatGPT	Generated the front, little-finger-side and thumb-side views and the terry and hoof-paint tiles.
Tripo	Generated the 3D base mesh and PBR maps from the four normalised views.
Other software	Blender 5.2.3 (bpy) for geometry, baking and rendering (Cycles); rembg/birefnet (segmentation); pyrender/trimesh (analysis renders); OpenCV/scipy/numpy/Pillow (image and mesh processing); unar; Chromium (screenshots, PDF).
Earlier windows	Slipper project's playbook and scripts (reused: kit generator structure, tile prep, bake methods, conventions).

9. Token accounting

[RECORDED] session counters (`get_session`, read 2026-11-01 01:35 UTC, i.e. before this report's work): input 5,716; output **461,480**; cache read 120,410,716; cache write 935,989; reported cost **USD 40.83**. Context window 1,000,000 tokens; auto-compaction threshold 784,000; one compaction (2026-10-28 23:33:32 UTC).

[RECORDED] transcript sums (364 API requests, final value per request, up to the moment of measurement): input 730; output **505,730**; cache read 125,888,687; cache write 1,416,842; total context processed 127,306,259. The two sources disagree (e.g. output 461k vs 506k). The reason is [UNKNOWN]; a plausible cause is that the session counter excludes requests from before worker restarts [INFERRED, unverified]. Treat the window total as **output ≈ 0.46–0.51 M tokens; input processed ≈ 121–127 M tokens (≈ 99 % cache reads)**.

[ESTIMATED] breakdown (method: characters measured in the transcript ÷ ~4 for prose, ÷ ~3.5 for code/JSON; images by Anthropic's documented sizing, long edge ≤ 1568 px, tokens ≈ w×h/750; each item counted once, ignoring the repeated cache reads that dominate the processed total):

Category	Measured	Estimated tokens
User-typed text (21 messages incl. this request; plus intake answers)	8,467 + ~600 characters	~2.3k
Harness compaction summary (machine-written, not the user)	15,270 characters	~3.8k
AI visible text (replies)	55,952 characters	~14k
AI thinking (only 30,728 characters stored)	by subtraction from recorded output	~290–330k
Tool inputs written by the AI (code, commands, file contents)	624,806 characters	~160–200k
Tool outputs returned (text)	622,156 characters	~150–180k
Uploaded images (9)	pixel dimensions	~13.8k
Images viewed by the AI (127 incl. repeats)	pixel dimensions	~154k
Attached files read as text (PLAYBOOK.md 20 KB, playbook scripts, docs)	included in tool outputs	(~5k for PLAYBOOK.md)
Generated images	none generated by this AI; renders enter only as viewed images above	0
Archives, STL, OBJ (binary)	processed by tools; only summaries entered the context	in tool outputs

10. Open questions and handoff

State at the end of production (2026-10-29 01:21 UTC) [RECORDED]: v4 final (20 mm smoothing, specks fixed), displacement variant, integration note; repo clean, last commit `9ce16ec` pushed; no pull request opened; latest handoff zips (five parts) delivered. Nothing awaited the AI.

Open / unresolved:

1. **Approvals not explicitly given:** the 100k budget, atlas layout, AO/gamma/terry-warming adjustments, the smoothing *method* (only the strength was chosen) [RECORDED: no explicit approval].
2. **Known limitations** (in the handoff): AO baked with the toes closed (spread pose slightly dark); flat inner walls read grey at grazing angles; thumb not separately animatable; cuff closed; hoof side profile, palm and thickness follow AI views; faint terry tile mirrored repeat; displacement fade band across the knuckle line on closed gloves; GLB uses 4K maps (drop to 2K for real-time).
3. **[UNKNOWN] for the master document:** ChatGPT model/version and the number of attempts for the side views and tiles; Tripo version/settings; the user's local time zone; whether `cowMask1.stl` is the 2014 MakerBot Digitizer scan (stated in the brief, not in the file).
4. **Repository nit:** `scripts/bl_split_hoof.py` exists both in `scripts/` and `scripts/superseded/` (a later bulk copy restored it) [RECORDED: file listing]; harmless.

5. **Next steps offered:** integrate via `INTEGRATION_blender_render.md`; optional spread-pose AO re-bake, 2K/50k real-time LOD, hollow cuff, shorter displacement fade.

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11. Figure register (visual materials)

203 figures, IDs **GLOVES-F001 … GLOVES-F203**, in chronological order. User-supplied inputs come first (F001–F029), then everything made or viewed in this window. Each figure's file in the package is named `<ID>_<slug>.<ext>`.

- **Viewed images:** every image the AI opened is included. When the file had been overwritten later, the copy stored in the transcript is used, at the resolution the AI saw. The provenance column of `figures.csv` says which.
- **Thumbnails:** the AI's 16 thumbnails of the reference photos (viewed 18:23–18:24) are merged into F001–F016.
- **Git history:** superseded repo versions were recovered from git.
- **Not in the register:** the slipper worked-example images inside the playbook zip (never opened here).
- **Re-sent copies:** JPEG copies of figures sent in zips appear separately only where the content differs.

Figure ID	Filename	Type	Who made it	What it shows	Step / round	Generation prompt (if AI-generated)	Recommended?
GLOVES-F001	GLOVES-F001_left_flatlay_D800.jpg	reference photo	user	Left glove laid flat on a black cutting mat beside the cow mask, back of the hand up, hoof toes parted (Nikon D800, 24 mm; 3159x3072). Source of REF_A, the front view edit, the scale measurement frame and the paint/terry swatches. Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	yes
GLOVES-F002	GLOVES-F002_costume_01.jpg	reference photo	user	Full costume (cow mask, plaid robe, pink udder) seated in an office chair, grey backdrop; both gloves on the armrests. Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	maybe
GLOVES-F003	GLOVES-F003_costume_02.jpg	reference photo	user	Costume seated, pink backdrop, one gloved hand raised to the face; hoof seen from the side. Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	maybe
GLOVES-F004	GLOVES-F004_costume_03.jpg	reference photo	user	Costume seated, pink backdrop, arm across the body; glove seen from above. Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	no
GLOVES-F005	GLOVES-F005_costume_04.jpg	reference photo	user	Costume seated, pink backdrop, both gloves resting on the thighs. Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	no
GLOVES-F006	GLOVES-F006_costume_05.jpg	reference photo	user	Costume seated holding a tabloid magazine and a snack bag; gloves gripping props (source of crop h). Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	no
GLOVES-F007	GLOVES-F007_costume_06.jpg	reference photo	user	Performer standing in the bodysuit with a striped jacket, mask off; gloves at the sides, green backdrop (source of crop c). Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	no
GLOVES-F008	GLOVES-F008_costume_07.jpg	reference photo	user	Performer reclining on the floor, mask off, one gloved hand raised with the hoof toes parted (source of REF_B / crop b). Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	yes
GLOVES-F009	GLOVES-F009_costume_08.jpg	reference photo	user	Performer reclining, mask off, red lingerie top; glove resting on the knee. Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	no
GLOVES-F010	GLOVES-F010_costume_09.jpg	reference photo	user	Costume reclining with mask on, pink backdrop, holding a printed sheet (source of crop d). Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	no
GLOVES-F011	GLOVES-F011_costume_10.jpg	reference photo	user	Mask close-up, teal backdrop, gloved hand holding a scroll tied with a lilac ribbon; hoof closed (source of REF_C after ribbon inpainting, DETAIL crops and the terry/paint swatch B). Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	yes
GLOVES-F012	GLOVES-F012_costume_11.jpg	reference photo	user	Same set-up as 10 of 13, slightly different pose; hoof closed holding the scroll. Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	maybe

Figure ID	Filename	Type	Who made it	What it shows	Step / round	Generation prompt (if AI-generated)	Recommended?
GLOVES-F013	GLOVES-F013_costume_12.jpg	reference photo	user	Costume with a police-costumed performer holding a baton; the cow holds a 'SAVE COWS SLAUGHTER HUMANS' cardboard sign. Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	no
GLOVES-F014	GLOVES-F014_costume_13.jpg	reference photo	user	Costume lying on red/yellow plastic sheeting, overhead view; one glove visible on the floor (source of crop g). Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	no
GLOVES-F015	GLOVES-F015_right_alt_6D.jpg	reference photo	user	Right glove on a collage of printed papers with a cyan hand-drawn ellipse; hoof looks glossy green-black (Canon EOS 6D, EF40mm). Not used for the cut-outs (did not segment cleanly). Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	no
GLOVES-F016	GLOVES-F016_right_flatlay.jpg	reference photo	user	Right glove laid flat on the black mat beside the mask (no EXIF camera data). Not used for the cut-outs. Viewed by the AI as a 1200 px thumbnail at 18:23-18:24.	S1 Intake/references	–	maybe
GLOVES-F017	GLOVES-F017_gloves_with_mask_photo1.jpg	reference photo	user	Both gloves laid beside the cow mask on the black mat (darker exposure). Hooves lost in the glove segmentation of this frame, so photo 2 was used for measuring.	S2 Scale	–	maybe
GLOVES-F018	GLOVES-F018_gloves_with_mask_photo2.jpg	reference photo	user	Same frame as photo 1, brighter; identified as the same Nikon D800 24 mm frame as the left flat-lay (template match 0.987); used to measure real glove size.	S2 Scale	–	yes
GLOVES-F019	GLOVES-F019_chatgpt_front_view.webp	AI-generated image	ChatGPT image generation (run by the user)	ChatGPT front view (back of the hand), hoof closed and joined as requested, thumb on the right; 1172x1832. Approved as 1_FRONT_approved.	S3 ChatGPT views	Kit STEP 2 prompt (verbatim in report Appendix A.2), pasted after STEP 1 (Appendix A.1); attachments REF_A, REF_C, DETAIL_3, DETAIL_4. Attempt: first image returned (user: 'the first image that came back').	yes
GLOVES-F020	GLOVES-F020_chatgpt_front_view_resent.webp	AI-generated image	ChatGPT image generation (run by the user)	Byte-identical re-send of F019 (MD5 f216b0c040...) attached with the two side views at 20:35.	S3 ChatGPT views	Kit STEP 2 prompt (verbatim in report Appendix A.2), pasted after STEP 1 (Appendix A.1); attachments REF_A, REF_C, DETAIL_3, DETAIL_4	no
GLOVES-F021	GLOVES-F021_chatgpt_side_little_finger.webp	AI-generated image	ChatGPT image generation (run by the user)	ChatGPT little-finger side view: slim profile, one black hoof wedge on top, no thumb; 1254x1254. Approved.	S3 ChatGPT views	Kit STEP 3 prompt (Appendix A.3); attachments 1_FRONT_approved, REF_B, DETAIL_2. Attempts: [UNKNOWN].	yes
GLOVES-F022	GLOVES-F022_chatgpt_side_thumb.webp	AI-generated image	ChatGPT image generation (run by the user)	ChatGPT thumb side view: thumb toward the camera, plain terry; 1254x1254. Approved.	S3 ChatGPT views	Kit STEP 4 prompt (Appendix A.4); attachments 1_FRONT_approved, 2_SIDE_LITTLE_FINGER_approved, REF_A. Attempts: [UNKNOWN].	yes
GLOVES-F023	GLOVES-F023_user_hand_split_reference.jpg	reference photo	user	The user's own hand making a Vulcan-salute split between the middle and ring fingers, sent to show where the glove must part.	S7 Split round 2	–	yes
GLOVES-F024	GLOVES-F024_chatgpt_tile_terry.webp	texture (AI-generated)	ChatGPT image generation (run by the user)	ChatGPT terry tile: dense cream cotton loops, flat light; 2000x2000 (the chat attachment was downscaled to 1932 px).	S11 Tiles	TILE 1 prompt (Appendix A.6); attachments ref_a_terry_loops.jpg, ref_b_terry_closeup.jpg. Attempts: [UNKNOWN].	yes

Figure ID	Filename	Type	Who made it	What it shows	Step / round	Generation prompt (if AI-generated)	Recommended?
GLOVES-F025	GLOVES-F025_chatgpt_tile_hoof_paint.webp	texture (AI-generated)	ChatGPT image generation (run by the user)	ChatGPT hoof-paint tile: black paint with long vertical brush strokes and grey drag streaks; 1254x1254.	S11 Tiles	TILE 2 prompt (Appendix A.6); attachments ref_a_brush_strokes.jpg, ref_b_satin_sheen.jpg. Attempts: [UNKNOWN].	yes
GLOVES-F026	GLOVES-F026_tripo_v1_basecolor_8k.jpg	texture	Tripo (run by the user)	Tripo v1 base-colour map, 8192x8192 JPEG, from the downloaded model.	S5 Tripo evaluation	Tripo multiview settings: [UNKNOWN]	maybe
GLOVES-F027	GLOVES-F027_tripo_v1_normal_4k.png	texture	Tripo (run by the user)	Tripo v1 normal map, 4096x4096 PNG.	S5 Tripo evaluation	–	no
GLOVES-F028	GLOVES-F028_tripo_v1_metallic_4k.png	texture	Tripo (run by the user)	Tripo v1 metallic map, 4096x4096 PNG; hoof mean metallic 0.923 (wrong for painted card; set to 0).	S5 Tripo evaluation	–	no
GLOVES-F029	GLOVES-F029_tripo_v1_roughness_4k.png	texture	Tripo (run by the user)	Tripo v1 roughness map, 4096x4096 PNG (did not survive the OBJ import into Blender).	S5 Tripo evaluation	–	no
GLOVES-F030	GLOVES-F030_crop_a_worn_ribbon.jpg	screenshot	this AI (Claude)	AI working crop: worn glove with hoof slightly parted and the lilac ribbon, from Original Costume 10 of 13 [INFERRED], assessed as a reference candidate.	S1 References	–	no
GLOVES-F031	GLOVES-F031_crop_b_raised_hand_V.jpg	screenshot	this AI (Claude)	AI working crop: raised worn glove with the toes parted in a V, green backdrop (Original Costume 7 of 13 [INFERRED]).	S1 References	–	no
GLOVES-F032	GLOVES-F032_crop_c_jacket.jpg	screenshot	this AI (Claude)	AI working crop: torso with striped jacket; glove too small to use (Original Costume 6 of 13).	S1 References	–	no
GLOVES-F033	GLOVES-F033_crop_d_sheet.jpg	screenshot	this AI (Claude)	AI working crop: closed hoof holding a printed sheet (Original Costume 9 of 13).	S1 References	–	no
GLOVES-F034	GLOVES-F034_crop_e_flat_toes_rim.jpg	screenshot	this AI (Claude)	AI working crop: flat left glove's two toe caps and raised rim, close-up (later DETAIL_1).	S1 References	–	maybe
GLOVES-F035	GLOVES-F035_crop_g_floor.jpg	screenshot	this AI (Claude)	AI working crop: glove on red/yellow sheeting (Original Costume 13 of 13).	S1 References	–	no
GLOVES-F036	GLOVES-F036_crop_h_snack.jpg	screenshot	this AI (Claude)	AI working crop: glove holding a snack bag (Original Costume 5 of 13).	S1 References	–	no
GLOVES-F037	GLOVES-F037_crop_i_terry_dirt.jpg	screenshot	this AI (Claude)	AI working crop: terry of the flat left glove with a grey dirt mark (basis for the 'ignore dirt' instruction).	S1 References	–	no
GLOVES-F038	GLOVES-F038_crop_j_alt_image.jpg	screenshot	this AI (Claude)	AI working crop of the 'Alt Image' right glove (glossy green-black hoof, cyan annotation).	S1 References	–	no
GLOVES-F039	GLOVES-F039_seg_test_isnet_failed.jpg	diagram	this AI (Claude)	Background-removal test on the flat-lay (magenta = removed): the model kept only the black hooves and dropped the cream terry.	S1 References / failure	–	yes
GLOVES-F040	GLOVES-F040_seg_test_2.jpg	diagram	this AI (Claude)	Second segmentation test on the flat-lay: still only the hooves retained.	S1 References / failure	–	no
GLOVES-F041	GLOVES-F041_seg_test_birefnet_ok.jpg	diagram	this AI (Claude)	Segmentation with rembg birefnet-general: whole glove (terry + hooves) retained on magenta.	S1 References / fix	–	maybe
GLOVES-F042	GLOVES-F042_refs_sheet_v1_notch.jpg	comparison sheet	this AI (Claude)	First reference cut-out sheet (as viewed; file later overwritten).	S1 References	–	no
GLOVES-F043	GLOVES-F043_refs_sheet_v2.jpg	comparison sheet	this AI (Claude)	Reference cut-out sheet, second version.	S1 References	–	no
GLOVES-F044	GLOVES-F044_REF_B_C_ribbon_notch.jpg	comparison sheet	this AI (Claude)	REF_B and REF_C cut-outs; removing the lilac ribbon after segmentation cut a false notch into the top hoof toe.	S1 References / failure	–	yes
GLOVES-F045	GLOVES-F045_REF_B_after_inpaint.jpg	comparison sheet	this AI (Claude)	Cut-out after inpainting the ribbon before segmentation: clean hoof outline.	S1 References / fix	–	maybe
GLOVES-F046	GLOVES-F046_grid_REF_A.jpg	diagram	this AI (Claude)	REF_A with a coordinate grid, used to place the design-key badges.	S1 Design key	–	no
GLOVES-F047	GLOVES-F047_grid_REF_B.jpg	diagram	this AI (Claude)	REF_B with a coordinate grid.	S1 Design key	–	no

Figure ID	Filename	Type	Who made it	What it shows	Step / round	Generation prompt (if AI-generated)	Recommended?
GLOVES-F048	GLOVES-F048_grid_REF_C.jpg	diagram	this AI (Claude)	REF_C with a coordinate grid.	S1 Design key	–	no
GLOVES-F049	GLOVES-F049_REF_ABC_final.jpg	comparison sheet	this AI (Claude)	The three final reference cut-outs side by side: A flat left glove, B worn toes slightly parted, C worn hoof closed.	S1 References	–	yes
GLOVES-F050	GLOVES-F050_design_key_preview.jpg	diagram	this AI (Claude)	Preview of the annotated design key (numbered features on REF_A-C).	S1 Design key	–	no
GLOVES-F051	GLOVES-F051_design_key.png	diagram	this AI (Claude)	Annotated design key, 2860x1650: three reference cut-outs with 10 numbered features (thumb has no hoof, exactly two toe caps, satin paint, closed cleft, etc.). Sent 19:08 and attached in kit steps 1 and fix-ups.	S1 Design key	–	yes
GLOVES-F052	GLOVES-F052_REF_A_cutout.png	reference photo (processed)	this AI (Claude)	REF_A: white-background cut-out of the left flat-lay (1536 px).	S1 References	–	maybe
GLOVES-F053	GLOVES-F053_REF_B_cutout.png	reference photo (processed)	this AI (Claude)	REF_B: cut-out of the worn glove, toes slightly parted (1536 px).	S1 References	–	maybe
GLOVES-F054	GLOVES-F054_REF_C_cutout.png	reference photo (processed)	this AI (Claude)	REF_C: cut-out of the worn glove, hoof closed, ribbon inpainted (1536 px).	S1 References	–	maybe
GLOVES-F055	GLOVES-F055_DETAIL_1.jpg	reference photo (processed)	this AI (Claude)	DETAIL_1 (1024 px): toe-cap shape and raised rim.	S1 References	–	no
GLOVES-F056	GLOVES-F056_DETAIL_2.jpg	reference photo (processed)	this AI (Claude)	DETAIL_2 (1024 px): closed hoof seen from its tip, cleft across it.	S1 References	–	no
GLOVES-F057	GLOVES-F057_DETAIL_3.jpg	reference photo (processed)	this AI (Claude)	DETAIL_3 (1024 px): terry loop texture.	S1 References	–	no
GLOVES-F058	GLOVES-F058_DETAIL_4.jpg	reference photo (processed)	this AI (Claude)	DETAIL_4 (1024 px): hoof paint, brush streaks and rim.	S1 References	–	no
GLOVES-F059	GLOVES-F059_DETAIL_5.jpg	reference photo (processed)	this AI (Claude)	DETAIL_5 (1024 px): closed hoof from above, cleft line.	S1 References	–	no
GLOVES-F060	GLOVES-F060_guide_page_screenshot.png	screenshot	this AI (Claude)	Screenshot of the published 'Hoof Glove View Kit' guide page (artifact) used to check layout before publishing.	S1 Prompt kit	–	maybe
GLOVES-F061	GLOVES-F061_mask_scan_views.png	render	this AI (Claude)	Four renders of cowMask1.stl: the mask is a relief on a rectangular base block (print-base-like), not a full head.	S2 Scale	–	yes
GLOVES-F062	GLOVES-F062_mask_scan_top.png	render	this AI (Claude)	Top-surface renders of the mask scan from four angles.	S2 Scale	–	no
GLOVES-F063	GLOVES-F063_mask_heightmap.png	diagram	this AI (Claude)	Colour height map of the scan's top surface, used to locate the bead eyes.	S2 Scale	–	maybe
GLOVES-F064	GLOVES-F064_mask_heightmap_eyes.png	diagram	this AI (Claude)	Height-map crop around the two bead eyes.	S2 Scale	–	no
GLOVES-F065	GLOVES-F065_mask_rotated.png	render	this AI (Claude)	Mask scan re-oriented for eye identification.	S2 Scale	–	no
GLOVES-F066	GLOVES-F066_photo2_eye_detection.png	diagram	this AI (Claude)	Photo 2 crop with the yellow bead-eye colour detection (left bead leaked into warm-lit face).	S2 Scale	–	no
GLOVES-F067	GLOVES-F067_photo2_eye_circlefit.png	diagram	this AI (Claude)	Circle fits to the two bead eyes in photo 2.	S2 Scale	–	maybe
GLOVES-F068	GLOVES-F068_photo1_eye_circlefit.png	diagram	this AI (Claude)	Circle fits to the two bead eyes in photo 1.	S2 Scale	–	no
GLOVES-F069	GLOVES-F069_scan_eye_hough.png	diagram	this AI (Claude)	Hough-circle check of the bead eyes on the scan height map [INFERRED from filename].	S2 Scale	–	no
GLOVES-F070	GLOVES-F070_glove_segmentation_photos12.jpg	diagram	this AI (Claude)	birefnet glove masks on photos 1 and 2 (left glove red, right glove blue); photo 1 lost both hooves.	S2 Scale	–	maybe
GLOVES-F071	GLOVES-F071_photo2_axes.jpg	diagram	this AI (Claude)	Photo 2 with measurement axes drawn along each glove (cuff to hoof tip).	S2 Scale	–	maybe

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GLOVES-F072	GLOVES-F072_photo2_grid_right.png	diagram	this AI (Claude)	Gridded zoom of the right glove in photo 2 for picking landmarks.	S2 Scale	–	no
GLOVES-F073	GLOVES-F073_photo2_grid_left.png	diagram	this AI (Claude)	Gridded zoom of the left glove in photo 2 for picking landmarks.	S2 Scale	–	no
GLOVES-F074	GLOVES-F074_scale_annotation_v1_bug.jpg	diagram	this AI (Claude)	First annotated measurement image (as viewed): eye line drawn in the wrong place (crop offset omitted when drawing; numbers unaffected).	S2 Scale / failure	–	no
GLOVES-F075	GLOVES-F075_eyeline_fixed.jpg	diagram	this AI (Claude)	Crop confirming the corrected eye line lands on the beads.	S2 Scale / fix	–	no
GLOVES-F076	GLOVES-F076_scale_measured_photo2.jpg	diagram	this AI (Claude)	Measurement annotation on photo 2: bead-eye spacing, camera height ~44 cm, glove lengths 24.7 cm, terry 15.8/16.0 cm, toe caps 8.3-9.4 cm.	S2 Scale	–	yes
GLOVES-F077	GLOVES-F077_tripo_inputs_lineup.jpg	comparison sheet	this AI (Claude)	The four normalised Tripo inputs side by side on 1536 px squares with common top/bottom lines and centre axis.	S4 Tripo inputs	–	yes
GLOVES-F078	GLOVES-F078_tripo_input_front.png	AI-generated image (normalised)	this AI (Claude)	Tripo Front slot: approved front view normalised to 1536x1536 white square.	S4 Tripo inputs	–	maybe
GLOVES-F079	GLOVES-F079_tripo_input_back.png	AI-generated image (normalised)	this AI (Claude)	Tripo Back slot: the front view mirrored (palm), 1536x1536.	S4 Tripo inputs	–	no
GLOVES-F080	GLOVES-F080_tripo_input_left_thumb.png	AI-generated image (normalised)	this AI (Claude)	Tripo Left slot (object's own left = thumb side), 1536x1536.	S4 Tripo inputs	–	no
GLOVES-F081	GLOVES-F081_tripo_input_right_little.png	AI-generated image (normalised)	this AI (Claude)	Tripo Right slot (little-finger side), 1536x1536.	S4 Tripo inputs	–	no
GLOVES-F082	GLOVES-F082_tripo_inputs_lineup_sent.jpg	comparison sheet	this AI (Claude)	Line-up check as sent to the user (final file).	S4 Tripo inputs	–	no
GLOVES-F083	GLOVES-F083_tripo_v1_turnaround_textured.jpg	render	this AI (Claude)	Tripo v1 model, textured turnaround (Tripo's own texture).	S5 Tripo evaluation	–	yes
GLOVES-F084	GLOVES-F084_tripo_v1_turnaround_clay.jpg	render	this AI (Claude)	Tripo v1 model, clay turnaround (shape only).	S5 Tripo evaluation	–	yes
GLOVES-F085	GLOVES-F085_tripo_v1_silhouette_check.jpg	comparison sheet	this AI (Claude)	Tripo v1 silhouettes over the four input views (IoU 0.92-0.94).	S5 Tripo evaluation	–	yes
GLOVES-F086	GLOVES-F086_tripo_v1_closeups_defects.jpg	render	this AI (Claude)	Tripo v1 close-ups: invented bead seams down the sides and around the cuff, closed cuff cap, hoof rim and brush grooves.	S5 Tripo evaluation	–	yes
GLOVES-F087	GLOVES-F087_highres_inspection_a.png	render	this AI (Claude)	Renders of the welded full-detail mesh for seam inspection (as viewed; file later overwritten) [INFERRED purpose].	S6 Cleanup	–	no
GLOVES-F088	GLOVES-F088_highres_inspection_b.png	render	this AI (Claude)	Second version of the same inspection renders.	S6 Cleanup	–	no
GLOVES-F089	GLOVES-F089_seam_flatten_before_after_v1.jpg	comparison sheet	this AI (Claude)	First before/after of the bead-seam flattening (as viewed; superseded).	S6 Cleanup	–	no
GLOVES-F090	GLOVES-F090_seam_flatten_before_after.jpg	comparison sheet	this AI (Claude)	Bead seams before (top) and after (bottom) flattening; terry only, hoof rim kept; max move 1.4 mm.	S6 Cleanup	–	yes
GLOVES-F091	GLOVES-F091_split_r1_first_attempt.jpg	render	this AI (Claude)	Hoof-only split, first attempt (as viewed): a gap where the toes meet the mitten extension.	S7 Split round 1 / failure	–	no
GLOVES-F092	GLOVES-F092_split_r1_parts_separated.jpg	render	this AI (Claude)	Mitten and two toes rendered separately to locate the missing area.	S7 Split round 1	–	no

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GLOVES-F093	GLOVES-F093_split_r1_hoof_only_closed_spread.jpg	render	this AI (Claude)	Split round 1 (superseded): hooves only, pivot at the hoof base; closed and spread 18 deg (colours mark parts).	S7 Split round 1	–	yes
GLOVES-F094	GLOVES-F094_split_r1_sent_file.jpg	render	this AI (Claude)	Round-1 split image as sent (later deleted from the repo; recovered from git 69828a8).	S7 Split round 1	–	no
GLOVES-F095	GLOVES-F095_split_r2_attempt1.jpg	render	this AI (Claude)	Finger-pair split, first attempt (as viewed): palm stub poking out between the toes.	S7 Split round 2 / failure	–	maybe
GLOVES-F096	GLOVES-F096_split_r2_attempt2.jpg	render	this AI (Claude)	Finger-pair split, second attempt (as viewed).	S7 Split round 2	–	no
GLOVES-F097	GLOVES-F097_split_r2_final_closed_spread.jpg	render	this AI (Claude)	Final split: two finger pairs parting at the middle/ring web (pivot 13.9 cm above cuff), closed and spread 18 deg each way.	S7 Split round 2	–	yes
GLOVES-F098	GLOVES-F098_decimation_150k_100k_64k.jpg	comparison sheet	this AI (Claude)	Decimation comparison: full mesh vs 150k / 100k / 64k triangles per glove (close-ups).	S8 Optimise	–	yes
GLOVES-F099	GLOVES-F099_uv_r1_smartproject_66.png	diagram	this AI (Claude)	UV round 1: plain smart-project at 66 deg (~1,800 islands).	S9 UV / failure	–	maybe
GLOVES-F100	GLOVES-F100_uv_r2_smartproject_80.png	diagram	this AI (Claude)	UV round 2: smart-project at 80 deg.	S9 UV	–	no
GLOVES-F101	GLOVES-F101_bake_test_sheet_specks.jpg	render	this AI (Claude)	1K test bake: black/white specks from sub-texel sliver islands.	S9 Bake / failure	–	maybe
GLOVES-F102	GLOVES-F102_bake_test_preview.png	render	this AI (Claude)	Preview render of the 1K test bake.	S9 Bake / failure	–	no
GLOVES-F103	GLOVES-F103_uv_r3_smoothed_proxies.png	diagram	this AI (Claude)	UV round 3: smart-project on smoothed proxy meshes.	S9 UV	–	no
GLOVES-F104	GLOVES-F104_bake_test_zoom_specks.png	render	this AI (Claude)	Zoom on the speck artefacts.	S9 Bake / failure	–	no
GLOVES-F105	GLOVES-F105_uv_r4_chart_merged.png	diagram	this AI (Claude)	UV round 4 (adopted): islands < 40 faces merged, angle-based unwrap per chart: 64 islands, 63.9 % coverage.	S9 UV / fix	–	yes
GLOVES-F106	GLOVES-F106_bake_test2_clean.png	render	this AI (Claude)	Test bake after chart merging: no specks.	S9 Bake / fix	–	no
GLOVES-F107	GLOVES-F107_bake_4k_preview.png	render	this AI (Claude)	4K selected-to-active bake preview (colour + normal from the full mesh).	S9 Bake	–	maybe
GLOVES-F108	GLOVES-F108_pair_v1_render_a.jpg	render	this AI (Claude)	v1 pair render, first iteration (as viewed).	S10 v1 delivery	–	no
GLOVES-F109	GLOVES-F109_pair_v1_render_b.jpg	render	this AI (Claude)	v1 pair render, second iteration (as viewed).	S10 v1 delivery	–	no
GLOVES-F110	GLOVES-F110_spread_inner_wall_check.png	render	this AI (Claude)	Close-up of a spread finger pair's flat inner wall (reads grey at grazing angles).	S10 v1 delivery	–	no
GLOVES-F111	GLOVES-F111_pair_v1_render_final.png	render	this AI (Claude)	v1 pair: left glove spread 15 deg, right closed; Tripo's cleaned texture (route 9a).	S10 v1 delivery	–	yes
GLOVES-F112	GLOVES-F112_uv_mistake_specks_doc.jpg	comparison sheet	this AI (Claude)	Documentation image of the sliver-island speck mistake.	S9 UV / failure	–	yes
GLOVES-F113	GLOVES-F113_baked_4k_check_doc.jpg	render	this AI (Claude)	4K bake check renders (documentation).	S9 Bake	–	no
GLOVES-F114	GLOVES-F114_inner_walls_doc.jpg	render	this AI (Claude)	Spread inner walls close-up (documentation).	S10 v1 delivery	–	no
GLOVES-F115	GLOVES-F115_uv_atlas_v1v2.jpg	diagram	this AI (Claude)	UV atlas layout used for v1/v2 (64 islands, 63.9 %).	S9 UV	–	maybe
GLOVES-F116	GLOVES-F116_v1_tex_color.jpg	texture	this AI (Claude)	v1 baked base colour (Tripo's texture, cleaned), 4K.	S10 v1 delivery	–	no

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GLOVES-F117	GLOVES-F117_v1_tex_normal.jpg	texture	this AI (Claude)	v1 baked tangent normal, 4K.	S10 v1 delivery	–	no
GLOVES-F118	GLOVES-F118_v1_tex_roughness.jpg	texture	this AI (Claude)	v1 roughness derived from colour (terry 0.90, paint 0.55), 2K.	S10 v1 delivery	–	no
GLOVES-F119	GLOVES-F119_swatch_paint_p1.jpg	texture	this AI (Claude)	Candidate hoof-paint swatch 1 (rotated so strokes run vertically).	S11 Tile kit	–	no
GLOVES-F120	GLOVES-F120_swatch_paint_p3.jpg	texture	this AI (Claude)	Candidate hoof-paint swatch 3.	S11 Tile kit	–	no
GLOVES-F121	GLOVES-F121_swatch_paint_p2.jpg	texture	this AI (Claude)	Candidate hoof-paint swatch 2 (still diagonal at first).	S11 Tile kit	–	no
GLOVES-F122	GLOVES-F122_swatch_terry_t1.jpg	texture	this AI (Claude)	Candidate terry swatch.	S11 Tile kit	–	no
GLOVES-F123	GLOVES-F123_swatch_sheet_v1.jpg	comparison sheet	this AI (Claude)	Swatch sheet, first version (as viewed): second paint swatch diagonal.	S11 Tile kit	–	no
GLOVES-F124	GLOVES-F124_swatch_sheet_paint.jpg	comparison sheet	this AI (Claude)	Corrected paint swatch sheet: strokes vertical.	S11 Tile kit	–	maybe
GLOVES-F125	GLOVES-F125_swatch_sheet_terry.jpg	comparison sheet	this AI (Claude)	Terry swatch sheet.	S11 Tile kit	–	no
GLOVES-F126	GLOVES-F126_region_map_v1.jpg	diagram	this AI (Claude)	Material region map, first version (as viewed).	S11 Region map	–	no
GLOVES-F127	GLOVES-F127_region_map.png	diagram	this AI (Claude)	Approved region map: red = painted hoof (finger-pair texels above the hoof base plane), cream = terry.	S11 Region map	–	yes
GLOVES-F128	GLOVES-F128_tilekit_terry_ref_a.jpg	reference photo (processed)	this AI (Claude)	Tile kit attachment: terry loops swatch from Original Costume 10 of 13 (1024 px).	S11 Tile kit	–	maybe
GLOVES-F129	GLOVES-F129_tilekit_terry_ref_b.jpg	reference photo (processed)	this AI (Claude)	Tile kit attachment: closer terry swatch (1024 px).	S11 Tile kit	–	no
GLOVES-F130	GLOVES-F130_tilekit_paint_ref_a.jpg	reference photo (processed)	this AI (Claude)	Tile kit attachment: brush-stroke swatch from the left flat-lay, rotated 50 deg CCW so strokes run vertically.	S11 Tile kit	–	maybe
GLOVES-F131	GLOVES-F131_tilekit_paint_ref_b.jpg	reference photo (processed)	this AI (Claude)	Tile kit attachment: satin sheen swatch from Original Costume 10 of 13, rotated 125 deg CCW.	S11 Tile kit	–	no
GLOVES-F132	GLOVES-F132_mask_paint_4k.png	texture	this AI (Claude)	Paint/terry mask in the UV atlas (4K, lossless): white = paint, from geometry.	S11 Region map	–	no
GLOVES-F133	GLOVES-F133_tiling_check_v2_first.jpg	texture	this AI (Claude)	2x2 tiling check of the prepared terry and paint tiles (first preparation, as viewed).	S12 Material	–	maybe
GLOVES-F134	GLOVES-F134_live_material_first_check.png	render	this AI (Claude)	First live-material close-ups (whole glove, hoof strokes, terry, spread).	S12 Material	–	no
GLOVES-F135	GLOVES-F135_pair_v2_render_a.jpg	render	this AI (Claude)	v2 pair render, first iteration (as viewed): paint read charcoal, AO greyed the terry.	S12 Material / failure	–	no
GLOVES-F136	GLOVES-F136_pair_v2_render_b.png	render	this AI (Claude)	v2 pair render after an AO change ('barely changed').	S12 Material	–	no
GLOVES-F137	GLOVES-F137_live_vs_baked_v2.jpg	comparison sheet	this AI (Claude)	Live material (top) vs baked maps (bottom) from the same cameras, v2.	S12 Material	–	maybe
GLOVES-F138	GLOVES-F138_moved_copy_test.png	render	this AI (Claude)	Render from a moved copy of the live .blend (relative texture paths test).	S12 Packaging	–	no
GLOVES-F139	GLOVES-F139_final_pair_v2.png	render	this AI (Claude)	v2 final pair (slipper look): left spread, right closed.	S12 v2 delivery	–	yes
GLOVES-F140	GLOVES-F140_final_closeups_v2.png	render	this AI (Claude)	v2 close-ups: glove, hoof strokes, terry, spread.	S12 v2 delivery	–	yes

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GLOVES-F141	GLOVES-F141_v2_tile_paint_prepared.jpg	texture	this AI (Claude)	Prepared paint tile v2 (vertical seam fixed, gamma 1.3), 2048 px; still contains the lint flecks.	S12 Material	–	maybe
GLOVES-F142	GLOVES-F142_v2_height_paint.jpg	texture	this AI (Claude)	Paint height map v2 (dark = raised), 2048 px.	S12 Material	–	no
GLOVES-F143	GLOVES-F143_v2_rough_paint.jpg	texture	this AI (Claude)	Paint roughness v2 (0.50 black - 0.72 grey streaks), 2048 px.	S12 Material	–	no
GLOVES-F144	GLOVES-F144_v2_tiling_check.jpg	texture	this AI (Claude)	2x2 tiling check of the v2 prepared tiles.	S12 Material	–	no
GLOVES-F145	GLOVES-F145_tile_terry_prepared.jpg	texture	this AI (Claude)	Prepared terry tile (warmed, B x0.95), 2048 px; unchanged v2-v4.	S12 Material	–	maybe
GLOVES-F146	GLOVES-F146_height_terry.jpg	texture	this AI (Claude)	Terry height map (high-pass of luminance), 2048 px; used for bump and, softened, for displacement.	S12 Material	–	no
GLOVES-F147	GLOVES-F147_v2_tex_color.jpg	texture	this AI (Claude)	v2 final baked base colour (AO included), 4K.	S12 v2 delivery	–	maybe
GLOVES-F148	GLOVES-F148_v2_tex_normal.jpg	texture	this AI (Claude)	v2 final baked tangent normal, 4K.	S12 v2 delivery	–	no
GLOVES-F149	GLOVES-F149_v2_tex_roughness.jpg	texture	this AI (Claude)	v2 final baked roughness, 2K.	S12 v2 delivery	–	no
GLOVES-F150	GLOVES-F150_v2_tex_ao.jpg	texture	this AI (Claude)	v2 ambient occlusion (denoised, normalised by p90, gamma 0.8), 2K.	S12 v2 delivery	–	no
GLOVES-F151	GLOVES-F151_v2_tex_detail_normal.jpg	texture	this AI (Claude)	v2 detail normal baked from the full Tripo mesh (live material input), 4K.	S12 v2 delivery	–	no
GLOVES-F152	GLOVES-F152_v2_final_pair_jpg.jpg	render	this AI (Claude)	v2 final pair as delivered (JPEG of F-final_pair_v2).	S12 v2 delivery	–	no
GLOVES-F153	GLOVES-F153_v2_final_closeups_jpg.jpg	render	this AI (Claude)	v2 close-ups as delivered.	S12 v2 delivery	–	no
GLOVES-F154	GLOVES-F154_v2_check_live_baked_jpg.jpg	comparison sheet	this AI (Claude)	v2 live-vs-baked check as delivered.	S12 v2 delivery	–	no
GLOVES-F155	GLOVES-F155_smooth_perpart_clay_0_5_10.jpg	comparison sheet	this AI (Claude)	Clay: original vs per-part smoothing 5 mm vs 10 mm (whole glove).	S13 Smoothing round 1	–	maybe
GLOVES-F156	GLOVES-F156_smooth_perpart_clay_close.jpg	comparison sheet	this AI (Claude)	Clay close-up: original vs per-part 10 mm; toe terry band still lumpy.	S13 Smoothing round 1	–	no
GLOVES-F157	GLOVES-F157_smooth_perpart_clay_toepins.jpg	comparison sheet	this AI (Claude)	Clay: original vs per-part 10 mm with relaxed toe pins.	S13 Smoothing round 2	–	no
GLOVES-F158	GLOVES-F158_smooth_perpart_web_pits.jpg	comparison sheet	this AI (Claude)	Web close-ups, original (top) vs per-part 10 mm (bottom): deep pits above the web survive.	S13 Smoothing round 2 / failure	–	yes
GLOVES-F159	GLOVES-F159_smooth_unsplit_web.jpg	comparison sheet	this AI (Claude)	Web close-ups, original (top) vs unsplit 10 mm smoothing (bottom, full-res): pits gone, crease kept.	S13 Smoothing round 3 / fix	–	yes
GLOVES-F160	GLOVES-F160_before_after_v2_v3.jpg	comparison sheet	this AI (Claude)	Textured close-ups, v2 (top) vs v3 10 mm smoothing (bottom).	S13 v3 delivery	–	maybe
GLOVES-F161	GLOVES-F161_final_pair_v3.png	render	this AI (Claude)	v3 final pair (10 mm smoothing).	S13 v3 delivery	–	no
GLOVES-F162	GLOVES-F162_inner_wall_v2_v3.jpg	comparison sheet	this AI (Claude)	Spread toe inner wall, v2 (left) vs v3 (right): same grey grazing-angle look.	S13 v3 delivery	–	no
GLOVES-F163	GLOVES-F163_uv_atlas_v3.jpg	diagram	this AI (Claude)	UV atlas for v3 (58 islands, 65 %).	S13 v3 delivery	–	no
GLOVES-F164	GLOVES-F164_v3_tex_color.jpg	texture	this AI (Claude)	v3 baked base colour, 4K.	S13 v3 delivery	–	no
GLOVES-F165	GLOVES-F165_v3_tex_normal.jpg	texture	this AI (Claude)	v3 baked tangent normal, 4K.	S13 v3 delivery	–	no

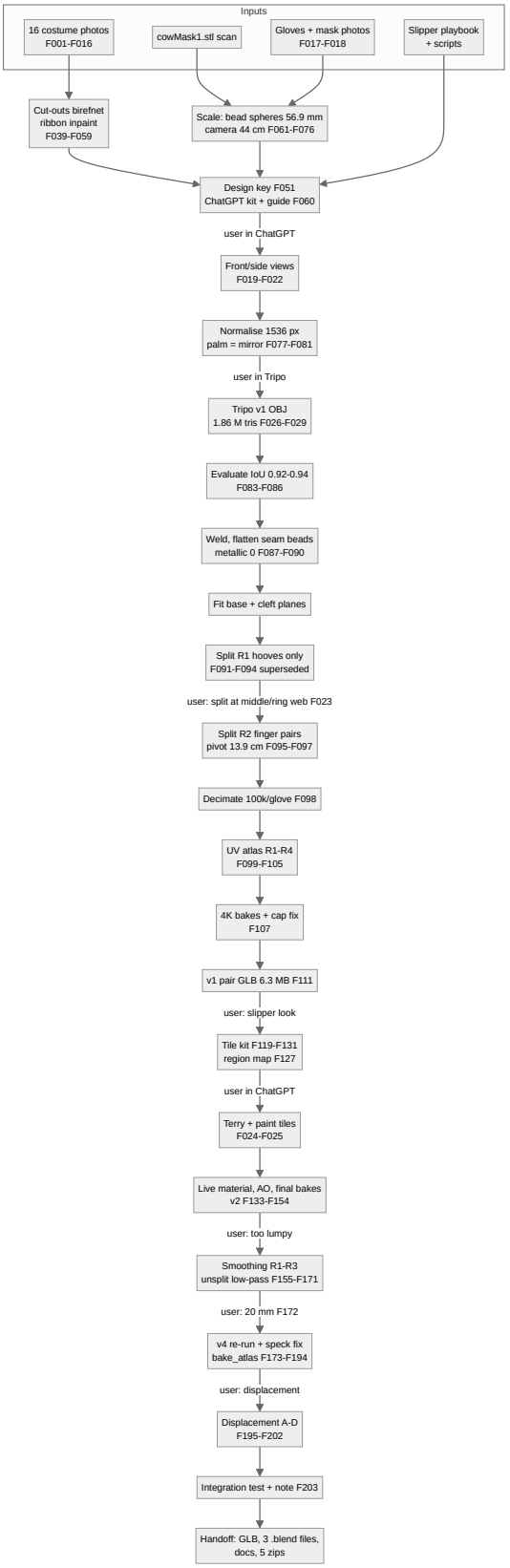
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GLOVES-F166	GLOVES-F166_v3_tex_roughness.jpg	texture	this AI (Claude)	v3 baked roughness, 2K.	S13 v3 delivery	–	no
GLOVES-F167	GLOVES-F167_v3_tex_ao.jpg	texture	this AI (Claude)	v3 ambient occlusion, 2K.	S13 v3 delivery	–	no
GLOVES-F168	GLOVES-F168_v3_tex_detail_normal.jpg	texture	this AI (Claude)	v3 detail normal (from the 10 mm smoothed full mesh), 4K.	S13 v3 delivery	–	no
GLOVES-F169	GLOVES-F169_v3_tex_mask_paint.png	texture	this AI (Claude)	v3 paint mask, 4K.	S13 v3 delivery	–	no
GLOVES-F170	GLOVES-F170_v3_final_closeups_jpg.jpg	render	this AI (Claude)	v3 close-ups as delivered.	S13 v3 delivery	–	no
GLOVES-F171	GLOVES-F171_v3_check_live_baked_jpg.jpg	comparison sheet	this AI (Claude)	v3 live-vs-baked check.	S13 v3 delivery	–	no
GLOVES-F172	GLOVES-F172_smoothing_10_20_30_clay.jpg	comparison sheet	this AI (Claude)	Clay comparison of 10 / 20 / 30 mm smoothing, back and thumb side (user chose 20 mm).	S14 Smoothing strength	–	yes
GLOVES-F173	GLOVES-F173_before_after_v4_specks_a.jpg	comparison sheet	this AI (Claude)	v2/v3/v4 close-ups, first v4 (as viewed): white specks on the hoof.	S14 v4 / failure	–	no
GLOVES-F174	GLOVES-F174_final_pair_v4_specks_a.jpg	render	this AI (Claude)	First v4 pair render (as viewed): white specks on the hoofs.	S14 v4 / failure	–	maybe
GLOVES-F175	GLOVES-F175_paint_tile_lint_flecks.png	texture	this AI (Claude)	Pixel crops of the ChatGPT paint tile showing cream lint-like flecks.	S14 Speck diagnosis	–	maybe
GLOVES-F176	GLOVES-F176_paint_tile_fleck_fix.png	texture	this AI (Claude)	Paint tile before (left) and after (right) inpainting the flecks.	S14 Speck diagnosis	–	no
GLOVES-F177	GLOVES-F177_before_after_v4_specks_b.jpg	comparison sheet	this AI (Claude)	v2/v3/v4 close-ups after the tile fix (as viewed): specks persist.	S14 v4 / failure	–	no
GLOVES-F178	GLOVES-F178_final_pair_v4_specks_b.jpg	render	this AI (Claude)	v4 pair after the tile fix (as viewed): specks persist.	S14 v4 / failure	–	no
GLOVES-F179	GLOVES-F179_speck_render_crops.png	render	this AI (Claude)	Pixel crops of the specks: cream terry-coloured patches on the paint (bake-margin overwrite).	S14 Speck diagnosis	–	yes
GLOVES-F180	GLOVES-F180_before_after_v2_v3_v4_final.jpg	comparison sheet	this AI (Claude)	Final v2 / v3 (10 mm) / v4 (20 mm) close-ups after the per-part bake compositing fix: no specks.	S14 v4 delivery	–	yes
GLOVES-F181	GLOVES-F181_final_pair_v4.png	render	this AI (Claude)	v4 final pair (20 mm smoothing, specks fixed).	S14 v4 delivery	–	yes
GLOVES-F182	GLOVES-F182_live_vs_baked_v4.jpg	comparison sheet	this AI (Claude)	Live material (top) vs baked v4 maps (bottom).	S14 v4 delivery	–	maybe
GLOVES-F183	GLOVES-F183_uv_atlas_v4.jpg	diagram	this AI (Claude)	UV atlas for v4 (76 islands, 64 %).	S14 v4 delivery	–	maybe
GLOVES-F184	GLOVES-F184_v4_tile_paint_prepared.jpg	texture	this AI (Claude)	Prepared paint tile v4: ~290 lint flecks inpainted, strokes and grey streaks kept, 2048 px.	S14 v4 delivery	–	yes
GLOVES-F185	GLOVES-F185_v4_height_paint.jpg	texture	this AI (Claude)	Paint height map v4, 2048 px.	S14 v4 delivery	–	no
GLOVES-F186	GLOVES-F186_v4_rough_paint.jpg	texture	this AI (Claude)	Paint roughness v4, 2048 px.	S14 v4 delivery	–	no
GLOVES-F187	GLOVES-F187_v4_tiling_check.jpg	texture	this AI (Claude)	2x2 tiling check of the v4 prepared tiles.	S14 v4 delivery	–	maybe
GLOVES-F188	GLOVES-F188_v4_tex_color.jpg	texture	this AI (Claude)	v4 final baked base colour (AO included), 4096x4096 JPEG; in the GLB.	S14 v4 delivery	–	yes
GLOVES-F189	GLOVES-F189_v4_tex_normal.jpg	texture	this AI (Claude)	v4 final baked tangent normal (OpenGL convention), 4096x4096; in the GLB.	S14 v4 delivery	–	maybe
GLOVES-F190	GLOVES-F190_v4_tex_roughness.jpg	texture	this AI (Claude)	v4 final baked roughness, 2048x2048.	S14 v4 delivery	–	no

Figure ID	Filename	Type	Who made it	What it shows	Step / round	Generation prompt (if AI-generated)	Recommended?
GLOVES-F191	GLOVES-F191_v4_tex_ao.jpg	texture	this AI (Claude)	v4 ambient occlusion, 2K.	S14 v4 delivery	–	no
GLOVES-F192	GLOVES-F192_v4_tex_detail_normal.jpg	texture	this AI (Claude)	v4 detail normal (from the 20 mm smoothed full mesh), 4K.	S14 v4 delivery	–	no
GLOVES-F193	GLOVES-F193_v4_tex_mask_paint.png	texture	this AI (Claude)	v4 paint mask, 4K.	S14 v4 delivery	–	no
GLOVES-F194	GLOVES-F194_v4_final_closeups_jpg.jpg	render	this AI (Claude)	v4 close-ups as delivered.	S14 v4 delivery	–	yes
GLOVES-F195	GLOVES-F195_displace_mask_preview.png	texture	this AI (Claude)	Terry displacement mask (UV atlas, 1024 preview): white = displaced, feathered to 0 near cut walls, 0 on the hoof.	S15 Displacement	–	maybe
GLOVES-F196	GLOVES-F196_bump_vs_displace_AB.jpg	comparison sheet	this AI (Claude)	A bump only vs B raw displacement 1.5 mm (as viewed).	S15 Displacement	–	no
GLOVES-F197	GLOVES-F197_bump_vs_displace_ABC.jpg	comparison sheet	this AI (Claude)	A / B / C (softened 0.3 mm) comparison (as viewed).	S15 Displacement	–	no
GLOVES-F198	GLOVES-F198_bump_vs_displace_closeups_ABCD.jpg	comparison sheet	this AI (Claude)	Raking-light and outline close-ups for A bump, B raw 1.5 mm, C softened 0.3 mm, D softened 0.12 mm (delivered).	S15 Displacement	–	yes
GLOVES-F199	GLOVES-F199_final_pair_displacement.png	render	this AI (Claude)	Pair rendered with terry displacement (setting D) from the repo copy of the .blend.	S15 Displacement	–	yes
GLOVES-F200	GLOVES-F200_displace_mask_4k.png	texture	this AI (Claude)	Terry displacement mask, 4096x4096 PNG (islands padded).	S15 Displacement	–	no
GLOVES-F201	GLOVES-F201_height_terry_displace.png	texture	this AI (Claude)	Terry height softened 0.12 mm for displacement, 2048 px.	S15 Displacement	–	no
GLOVES-F202	GLOVES-F202_bump_vs_displace_ABCD_full.jpg	comparison sheet	this AI (Claude)	Full 4x4 comparison sent to the user: rows A-D, columns raking light / outline / web spread / whole glove.	S15 Displacement	–	yes
GLOVES-F203	GLOVES-F203_integration_test_render.png	render	this AI (Claude)	Integration test: gloves appended into a fresh character file, parented to stand-in hand bones, rendered in Cycles.	S16 Integration note	–	maybe

Full metadata per figure (pixel size, bytes, MD5, source, provenance, whether and when it was sent to the user, when the AI first viewed it): `figures.csv`.

---

12. Pipeline diagram



Appendix A. Image-generation prompts (verbatim, as delivered in the kits)

The user pasted these into ChatGPT. The Step 1–4 text below is identical in the 19:27 build the user received and in the repo's final 20:36 build [RECORDED: git diff]. The README and checklists accompanying each step are included as they appeared in the zips.

A.1 STEP 1 (lock in the design; no image)

```
THE MISANTHROPIC COW HOOF GLOVE — FOUR-VIEW KIT
============================================================

Goal: four matching views of ONE left-hand glove (back of hand, little-finger side, thumb side, palm) for Tripo
multiview. Three are generated in ChatGPT; the palm view is a mirror of the back-of-hand view.
Work through STEP_1 to STEP_5 in order. Each package has a PROMPT.txt and the images to attach, numbered in attach
order.
Works in ChatGPT image generation or Gemini. Use the same chat throughout unless told otherwise.

GROUND RULES:
  - One view per message. Never ask for a sheet or grid of views.
  - In ChatGPT, use the Select tool on the image to brush over only the wrong area before sending a fix. Then only that
area is redrawn instead of the whole image.
  - If two fixes in a row make things worse, start a new chat: run Step 1 again, then attach your approved images and
carry on from where you were.
  - Approve a view only after checking it against the design key. One wrong view teaches Tripo the wrong shape.

DESIGN_KEY.png is your checklist for every result.

STEP 1 OF 5 — LOCK IN THE DESIGN
============================================================

Get the image model to read the glove correctly before it draws anything. No image is made in this step.

WHERE: Start a brand-new chat.

ATTACH THESE, IN THIS ORDER:
  1. 1_REF_A_left_glove_flat_back_of_hand.png  (Real left glove, empty, laid flat, hoof toes parted)
  2. 2_REF_B_worn_side_hoof_almost_closed.png  (Real glove worn, toes slightly parted)
  3. 3_REF_C_worn_hoof_closed_from_above.png  (Real glove worn, hoof CLOSED)
  4. 4_DETAIL_1_hoof_toes_shape_and_rim.jpg  (Toe shape and rim, close-up)
  5. 5_DESIGN_KEY.png  (Annotated design key)

THEN PASTE THIS PROMPT (everything between the lines):
------------------------------------------------------------
I'm making a 3D model of a costume glove belonging to my character, The Misanthropic Cow. Over the next few messages
I'll ask you for three product-reference views of ONE of these gloves (a left glove): the back of the hand, the little-
finger side and the thumb side. They will be fed into an image-to-3D tool, so being accurate to the real glove matters
far more than looking nice.

Attached, in order:
1. REF_A: the real left glove, empty and laid flat, seen from the back of the hand. The hoof toes are parted here; in
my views they must be closed.
2. REF_B: the real glove being worn, hoof toes slightly parted. Ignore the black background.
3. REF_C: the real glove being worn with the hoof CLOSED, the way I want it.
4. DETAIL_1: close-up of the two toe caps and their rim. Ignore the white fur sleeve: it belongs to the bodysuit, not
the glove.
5. DESIGN_KEY: my annotated breakdown of the parts. It is for understanding only. Never draw its numbers, circles,
lines or text.

DESIGN BRIEF — THE MISANTHROPIC COW HOOF GLOVE
(Authoritative. Follow it for every image in this chat.)

WHAT IT IS: ONE handmade costume glove for a LEFT hand: a cream terry-cloth mitten with a separate thumb, whose fingers
end in a hard black cow's CLOVEN HOOF made of exactly two toe caps. Not a pair. Not a five-finger glove. Not a furry
paw. Not a cartoon plush hoof.

POSE (every view): as if worn by an invisible hand. Plump and filled out, fingers straight and together, the two hoof
toes CLOSED. The glove stands upright: hoof at the TOP of the frame, cuff opening at the BOTTOM. The thumb angles out
about 25-30 degrees from the hand, with a clear gap between thumb and hand down to the thumb's base. No hand, wrist,
arm or skin is visible anywhere.

PROPORTIONS (measured from my photos against a 3D scan of the mask): about 25 cm from cuff to hoof tip and about 9 cm
wide across the hand, so about 2.8 times as tall as wide. Each hoof toe cap is about 9 cm long: the hoof is the top
~36% of the length and the terry the lower ~64% (about 16 cm). The thumb is short, about 4.5-5 cm, and leaves the hand
about halfway up the terry. From the side the glove is slim: about 4-5 cm thick at the knuckles.

PARTS, TOP TO BOTTOM:
1. HOOF (top ~36%): two hard, hollow black toe caps side by side, one over the index+middle fingers and one over the
ring+little fingers. Seen from the back of the hand, each toe has a straight inner edge along the centre line and a
curved outer edge, and its tip leans in toward the centre. CLOSED, the two toes touch along one straight cleft line
down the middle of the hoof and their tips meet in a single point, so together they make a rounded, pointed-arch
outline like a real cow's hoof.
2. HOOF FROM THE SIDE: from the little-finger or thumb edge, the hoof is a slim, rounded wedge, as thick as the hand at
its base, tapering to a blunt point at the tip.
```

```
3. HOOF FINISH: satin black paint with soft grey brush-streak highlights, slightly uneven like painted papier-mache. A
thin raised rim runs along the toe edges and catches light-grey highlights. Satin, not glossy; not metallic; no green
or rainbow sheen.
4. HOOF BASE: each toe cap starts at a clean, straight, slightly slanted edge where the terry tucks inside it.
5. FINGERS: just below the hoof, the fingers sit in two terry compartments, with a soft crease between them that
continues the cleft line a short way down the glove.
6. THUMB: a separate terry thumb with a rounded tip. Terry only: NO hoof, cap, claw or nail.
7. BODY: cream / off-white looped terry cloth, like a bath towel or a fluffy sock (small dense loops about 2-3 mm).
Plain: no pattern, no visible seams. Clean cream: ignore the grey dirt marks and coloured studio light in my photos.
8. CUFF: a plain, straight, open edge at the wrist, about as wide as the palm. No ribbing, no band, no fur trim.

NEVER ADD: five separate fingers, claws, talons or nails, a hoof or cap on the thumb, a third hoof toe, a horseshoe,
cow-print spots, a white fur cuff or sleeve, a knitted rib cuff, contrasting stitching, gloss, chrome or a green sheen
on the hoof, a hand, wrist, arm or skin, a second glove, a stand or mannequin, or any text, labels, numbers or arrows.

Don't make an image yet. First reply with:
(a) anything in the photos that contradicts or adds to the brief, and
(b) one or two sentences each on what the back-of-hand view, the little-finger-side view and the thumb-side view of
THIS glove should look like, with the hoof closed.
------------------------------------------------------------

BEFORE YOU APPROVE IT, CHECK:
  [ ] It says the hoof is exactly TWO toe caps (a cloven hoof), not five fingers or claws.
  [ ] It says the thumb is plain terry with NO hoof.
  [ ] It describes the hoof as CLOSED, with the toes meeting along a straight cleft and the tips meeting in one point.
  [ ] It calls the hoof satin black with grey brush highlights, not glossy.
  [ ] It does not add a fur cuff or sleeve.

NEXT: If it gets any of these wrong, correct it in one line (for example: "Correction: the thumb has no hoof; it is
plain terry.") and wait for it to confirm before moving on.

Something wrong? Open the FIX_UP_PROMPTS package and paste the matching fix.
```

A.2 STEP 2 (front view; produced F019)

```
STEP 2 OF 5 — FRONT VIEW: BACK OF THE HAND
============================================================

REF_A already shows this view of the real glove, so this is an edit, not an invention. This view sets the scale and
look for every other one.

WHERE: Same chat as Step 1.

ATTACH THESE, IN THIS ORDER:
  1. 1_REF_A_left_glove_flat_back_of_hand.png  (This exact view of the real glove (edit from it))
  2. 2_REF_C_worn_hoof_closed_from_above.png  (How the CLOSED hoof looks)
  3. 3_DETAIL_3_terry_loop_texture.jpg  (Terry loop texture)
  4. 4_DETAIL_4_hoof_surface_brush_texture.jpg  (Hoof paint, brush streaks and rim)

THEN PASTE THIS PROMPT (everything between the lines):
------------------------------------------------------------
IMAGE 1 OF 3: FRONT VIEW (BACK OF THE HAND).

Attached, in order: REF_A (the real glove from exactly this side; use it as the starting point), REF_C (how the CLOSED
hoof looks), DETAIL_3 (the terry loops), DETAIL_4 (the hoof's paint, brush streaks and rim).

Make a photograph of this exact glove (one LEFT glove) seen straight-on from the back of the hand, starting from REF_A,
with these changes only:
- It is worn by an invisible hand: plump and rounded, not flat and empty.
- The two hoof toes are CLOSED: they touch along one straight vertical cleft down the centre of the hoof, and their
tips meet in a single point at the top. The closed hoof has a rounded, pointed-arch outline.
- The terry is clean cream (no grey dirt marks).
- The glove stands perfectly upright, hoof at the top, cuff at the bottom; the hoof is symmetrical about its cleft.

What this view must show, top to bottom:
- The closed black cloven hoof (top ~36% of the height): two toe caps side by side, straight cleft down the middle,
satin black with soft grey brush highlights, thin rim along the edges.
- The clean straight edge where the terry tucks into the toe caps.
- The cream terry hand, with a soft crease continuing the cleft a short way down.
- The THUMB on the RIGHT side of the glove, angled out about 25-30 degrees, with white background visible between thumb
and hand. Plain terry, rounded tip, NO hoof.
- The straight open cuff edge at the bottom of the frame, about as wide as the palm.

CAMERA AND FRAMING: Camera at the glove's mid-height, looking straight across (level, not from above). Long-lens, near-
orthographic look with very little perspective distortion. Square 1:1 image. Pure white background. The glove is
centred, upright, and fills about 85% of the image height. Soft, even, diffuse studio light; no hard shadows and no
```

```
floor shadow. Photorealistic product photograph of the real handmade object, with its real terry and painted textures.
No text, labels, numbers, arrows, borders or watermark.

Follow the design brief exactly.
------------------------------------------------------------

BEFORE YOU APPROVE IT, CHECK:
  [ ] Thumb on the RIGHT, plain terry, no hoof; white gap between thumb and hand.
  [ ] Exactly two toe caps, CLOSED: one straight cleft down the middle, tips meeting in one point.
  [ ] Hoof is satin black with grey brush highlights, not glossy, metallic or green.
  [ ] Hoof is roughly the top third of the glove; glove about 2.8 times as tall as wide; short thumb.
  [ ] Upright, straight-on, one glove, no arm, no fur cuff, no text, white background.

SAVE THE APPROVED IMAGE AS: 1_FRONT_approved.png

NEXT: Download it and rename it 1_FRONT_approved.png. You'll attach it to every later step and mirror it in Step 5.

Something wrong? Open the FIX_UP_PROMPTS package and paste the matching fix.
```

A.3 STEP 3 (little-finger side; produced F021)

```
STEP 3 OF 5 — SIDE VIEW: LITTLE-FINGER EDGE
============================================================

The simplest side: the thumb is hidden behind the hand, so this is just the glove's slim profile and the side of the
hoof.

WHERE: Same chat. If the chat has started to drift, open a fresh one and run Step 1 again first.

ATTACH THESE, IN THIS ORDER:
  1. 1_FRONT_approved.png  <- YOUR approved image from an earlier step (not in this folder)
  2. 2_REF_B_worn_side_hoof_almost_closed.png  (Real glove worn: thickness of the toes)
  3. 3_DETAIL_2_closed_hoof_end_on_cleft.jpg  (Closed hoof seen from its tip end)

THEN PASTE THIS PROMPT (everything between the lines):
------------------------------------------------------------
IMAGE 2 OF 3: SIDE VIEW (LITTLE-FINGER EDGE).

Attached, in order: 1_FRONT_approved (my approved image 1; match it exactly), REF_B (the real glove worn, showing how
thick the toe caps are), DETAIL_2 (the closed hoof seen from its tip end: rounded, with the cleft across it).

Make a photograph of the same single glove, rotated exactly 90 degrees so the camera looks straight at its LITTLE-
FINGER EDGE. The glove is still upright: hoof at the top, cuff at the bottom. The back of the hand faces the RIGHT edge
of the frame and the palm faces the LEFT edge.

What this view must show:
- A slim, upright profile: the glove is only about 4-5 cm thick, about half as wide as in the front view.
- At the top, the side of the closed hoof: a slim, rounded black wedge, as thick as the hand at its base, tapering to a
blunt point at the tip. Only the outer side of ONE toe cap is visible; the cleft is on the far side and hidden. Same
satin black, grey brush highlights and thin rim.
- The clean straight edge where the terry tucks into the hoof.
- Below it, the cream terry hand and wrist in profile, slightly fuller toward the knuckles.
- The THUMB is completely hidden behind the hand: no thumb visible.
- The open cuff edge at the bottom, seen edge-on.

MATCH MY APPROVED VIEW: same glove, same scale, same camera height, same lighting, same white background, same terry
colour and hoof finish. The hoof tip and the bottom of the cuff sit at the same heights in the frame as in
1_FRONT_approved.

CAMERA AND FRAMING: Camera at the glove's mid-height, looking straight across (level, not from above). Long-lens, near-
orthographic look with very little perspective distortion. Square 1:1 image. Pure white background. The glove is
centred, upright, and fills about 85% of the image height. Soft, even, diffuse studio light; no hard shadows and no
floor shadow. Photorealistic product photograph of the real handmade object, with its real terry and painted textures.
No text, labels, numbers, arrows, borders or watermark.

Follow the design brief exactly.
------------------------------------------------------------

BEFORE YOU APPROVE IT, CHECK:
  [ ] True edge-on profile: much narrower than the front view.
  [ ] No thumb visible.
  [ ] Hoof side is one slim black wedge (no fingers, no claws), same finish as the front view.
  [ ] Hoof tip and cuff bottom at the same heights as in 1_FRONT_approved.
  [ ] One glove, no arm, no text, white background.

SAVE THE APPROVED IMAGE AS: 2_SIDE_LITTLE_FINGER_approved.png
```

```
NEXT: Save it as 2_SIDE_LITTLE_FINGER_approved.png and move on to Step 4.

Something wrong? Open the FIX_UP_PROMPTS package and paste the matching fix.
```

A.4 STEP 4 (thumb side; produced F022)

```
STEP 4 OF 5 — SIDE VIEW: THUMB EDGE
============================================================

The hardest view: the thumb comes toward the camera. It's optional. If it fights you after two fixes, skip it; Tripo
works without it.

WHERE: Same chat.

ATTACH THESE, IN THIS ORDER:
  1. 1_FRONT_approved.png  <- YOUR approved image from an earlier step (not in this folder)
  2. 2_SIDE_LITTLE_FINGER_approved.png  <- YOUR approved image from an earlier step (not in this folder)
  3. 3_REF_A_left_glove_flat_back_of_hand.png  (Real glove: thumb shape and position)

THEN PASTE THIS PROMPT (everything between the lines):
------------------------------------------------------------
IMAGE 3 OF 3: SIDE VIEW (THUMB EDGE).

Attached, in order: 1_FRONT_approved and 2_SIDE_LITTLE_FINGER_approved (my approved images; match them), REF_A (the
real glove; it shows the thumb's shape and where it joins the hand).

Make a photograph of the same single glove seen from the opposite side to image 2: the camera looks straight at its
THUMB EDGE. The glove is upright: hoof at the top, cuff at the bottom. The back of the hand faces the LEFT edge of the
frame and the palm faces the RIGHT edge.

What this view must show:
- The same slim profile as image 2, almost a mirror image of it: same height, same thickness, same hoof wedge at the
top (one toe cap's outer side; the cleft is hidden).
- The THUMB is on the near side, pointing up and toward the camera, so it looks short and rounded. It overlaps the
lower half of the hand, a little toward the palm side (right). Plain terry with a rounded tip: NO hoof, cap or nail.
- The cream terry and the open cuff edge at the bottom, as in image 2.

MATCH MY APPROVED VIEW: same glove, same scale, same camera height, same lighting, same white background, same terry
colour and hoof finish. The hoof tip and the bottom of the cuff sit at the same heights in the frame as in
1_FRONT_approved.

CAMERA AND FRAMING: Camera at the glove's mid-height, looking straight across (level, not from above). Long-lens, near-
orthographic look with very little perspective distortion. Square 1:1 image. Pure white background. The glove is
centred, upright, and fills about 85% of the image height. Soft, even, diffuse studio light; no hard shadows and no
floor shadow. Photorealistic product photograph of the real handmade object, with its real terry and painted textures.
No text, labels, numbers, arrows, borders or watermark.

Follow the design brief exactly.
------------------------------------------------------------

BEFORE YOU APPROVE IT, CHECK:
  [ ] Same slim profile, height and hoof as image 2 (roughly its mirror).
  [ ] Thumb visible on the near side, plain terry, no hoof.
  [ ] Still exactly one glove, upright, no arm.

SAVE THE APPROVED IMAGE AS: 3_SIDE_THUMB_approved.png

NEXT: Save it as 3_SIDE_THUMB_approved.png. If you skip it, just leave that Tripo slot empty.

Something wrong? Open the FIX_UP_PROMPTS package and paste the matching fix.
```

A.5 STEP 5 (palm view by mirroring; Tripo slots, final 20:36 version)

```
STEP 5 OF 5 — PALM VIEW (MIRROR) AND TRIPO UPLOAD
============================================================

No AI needed. The palm side of a mitten is the mirror image of the back of the hand, and a mirrored approved view is
better Tripo input than a freshly generated one.

WHERE: Any photo app, then Tripo.


DO THIS:
------------------------------------------------------------
```

```
MAKE THE PALM VIEW (no ChatGPT needed):
1. Open 1_FRONT_approved.png in any photo app.
2. Flip it horizontally (Mac Preview: Tools > Flip Horizontal; Windows Photos: Edit > Flip; Photoshop: Image > Image
Rotation > Flip Canvas Horizontal).
3. Save the copy as 4_BACK_PALM_mirrored.png. The thumb is now on the LEFT.

THEN SEND ME ALL THE IMAGES FIRST (recommended): attach 1_FRONT_approved, 2_SIDE_LITTLE_FINGER_approved,
3_SIDE_THUMB_approved (if you made it) and 4_BACK_PALM_mirrored here. I'll normalise them (same height, same centring,
white square) and give you back the exact files for Tripo.

Tripo slots are listed in TRIPO_UPLOAD.txt.
------------------------------------------------------------

BEFORE YOU APPROVE IT, CHECK:
  [ ] The mirrored copy has the thumb on the LEFT.
  [ ] All views show the hoof CLOSED and the same glove size.

SAVE THE APPROVED IMAGE AS: 4_BACK_PALM_mirrored.png

NEXT: Send the views back to me for normalising before you upload to Tripo.

Something wrong? Open the FIX_UP_PROMPTS package and paste the matching fix.

UPLOADING TO TRIPO (multiview)
============================================================

  Front  <- 1_FRONT_approved.png (back of hand, thumb on the right)
  Back   <- 4_BACK_PALM_mirrored.png (palm, thumb on the left)
  Left   <- 3_SIDE_THUMB_approved.png (thumb edge, the glove's own left; leave empty if you skipped Step 4)
  Right  <- 2_SIDE_LITTLE_FINGER_approved.png (little-finger edge, the glove's own right)

  - Send me the views first so I can normalise them; upload the normalised files I send back.
  - Tripo needs the front view plus at least one other. Leave out any view you don't trust.
  - Left/Right follow the slipper project (Left = the object's own left). If the model comes out twisted, or with a
thumb on both sides, swap the Left and Right images and run it again.
  - When it's done, download the model as a zip (OBJ + PBR textures) and upload it here.
```

A.6 Tile prompts (produced F024 and F025)

```
TEXTURE TILES FOR THE COW HOOF GLOVES — 2 images to make in ChatGPT
====================================================================
For each folder:
  1. Open a NEW ChatGPT chat (one chat per tile keeps them from blending together).
  2. Attach the two reference images in that folder.
  3. Paste that tile's prompt from PROMPTS.txt and send.
  4. Download the result and name it as shown (TILE_1_terry.png, TILE_2_hoof_paint.png).
  5. Send both tiles back to Claude.

What a good tile looks like:
  - The material fills the whole square, edge to edge. No objects, edges, rims, folds or seams.
  - Flat, even light. No shadows, shine spots or dark corners.
  - Same size of detail everywhere (no zooming in at the centre).
  - Hoof tile: the brush strokes run straight UP AND DOWN (top edge to bottom edge). That is the
    direction they will run on the hooves, from the base to the tip.
  - Don't worry if it isn't perfectly seamless. Claude fixes the seams.

The reference photos look greenish/grey because of the studio lighting. The prompts give the real colours.
The hoof's raised rim and the brush grooves already in the 3D shape are kept as they are; the tile adds the
painted surface on top.

Paste one prompt per new chat, with that folder's two reference images attached.

------------------------------------------------------------
TILE 1: CREAM TERRY CLOTH (save as TILE_1_terry.png)
Attach: 1_Terry/ref_a_terry_loops.jpg and 1_Terry/ref_b_terry_closeup.jpg
------------------------------------------------------------
Create a seamless, tileable square texture of the looped terry cloth in the attached photos (a cream terry glove, like
a bath towel or a fluffy sock).

Important: the 3D model already has the glove's soft lumps and folds as shapes. This texture is only the close-up
fabric surface: small, dense, curly cotton loops.
- Colour: clean warm cream / off-white (about #EDE6D2), with very subtle natural variation between loops. The photos
look greenish only because of the studio light. No dirt marks.
- View: straight down at a flat piece of terry, filling the entire square edge to edge.
- Scale: the square shows about 4 cm × 4 cm of fabric. Each loop is about 2-3 mm, densely packed, slightly tangled,
like the second photo.
```

```
- Lighting: completely flat and even. No shadows, highlights, vignetting or gradients.
- Nothing else: no seams, edges, folds, objects or text.
- It must tile seamlessly when repeated left-right and up-down.
- Square image, as large as you can make it.

------------------------------------------------------------
TILE 2: PAINTED HOOF WITH LONG BRUSH STROKES (save as TILE_2_hoof_paint.png)
Attach: 2_Hoof_paint/ref_a_brush_strokes.jpg and 2_Hoof_paint/ref_b_satin_sheen.jpg
------------------------------------------------------------
Create a seamless, tileable square texture of the painted surface in the attached photos: black acrylic paint brushed
by hand over papier-mache / cardboard (it's a costume cow hoof).

The most important feature is the LONG BRUSH STROKES:
- Long, parallel brush strokes running STRAIGHT UP AND DOWN, from the top edge of the square to the bottom edge, each
stroke continuous for most of the height. They are slightly wavy and vary a little in width, like the first photo.
- Where the brush dragged thin, soft grey streaks show along the strokes (thin paint over grey card). These grey
streaks are part of the paint, not reflections.
- Fine, faint bristle lines inside each stroke, also running up and down.
- Overall colour: deep satin black (about #1A1A1A) with the grey streaks (about #5A5A5A) covering roughly a quarter of
the area. A few tiny pale specks are fine.
- Finish: satin, not glossy, not metallic, no green or rainbow sheen.
- View: straight down at a flat painted surface, filling the entire square edge to edge.
- Scale: the square shows about 6 cm × 6 cm of the hoof.
- Lighting: completely flat and even. No shadows, shine spots, vignetting or gradients.
- Nothing else: no edges, rims, seams, cracks, objects or text.
- It must tile seamlessly when repeated left-right and up-down (the strokes continue across the top and bottom edges).
- Square image, as large as you can make it.

------------------------------------------------------------
IF A RESULT IS WRONG (reply in the same chat):
- Strokes run sideways or diagonally: "Rotate the brush strokes so they run straight up and down, from the top edge to
the bottom edge. Keep everything else the same."
- Strokes too short or blotchy: "Make the brush strokes long and continuous, running the full height of the square,
like the first photo."
- Too glossy or shows reflections: "Make the lighting completely flat and even, like a scanned swatch. Satin finish, no
shine. Keep the strokes the same."
- Zoomed in at the centre or uneven: "Keep the same size of detail across the whole square, like a uniform material
swatch."
- Shows objects, edges, a rim or folds: "Only the flat material, filling the whole square edge to edge."
- Terry looks like fur or fleece: "These are small curly cotton loops like a bath towel, 2-3 mm each, not long fur or
fleece. Keep the colour the same."
- Terry looks dirty, grey or green: "Clean warm cream colour (#EDE6D2), no dirt marks. Keep the loops the same."
```

A.7 Fix-up prompts (available; whether any were used is [UNKNOWN])

```
FIX-UP PROMPTS
============================================================

Paste the matching fix in the same chat. Attach the detail image noted beside the fix if the model needs reminding.
In ChatGPT, brush over just the wrong area with the Select tool first, so only that part is redrawn.

Fix A — Hoof became fingers, claws or more than two toes
------------------------------------------------------------
The hoof is wrong. This glove ends in a cow's cloven hoof: exactly TWO hard black toe caps side by side, one over the
index+middle fingers and one over the ring+little fingers. No separate fingers, claws or nails (see DESIGN_KEY item 4).
Keep everything else in the image exactly the same and fix only the hoof.

Fix B — Hoof is open instead of closed
------------------------------------------------------------
Close the hoof: the two toe caps must touch along one straight cleft line down the middle, with their tips meeting in a
single point, like REF_C. Keep everything else exactly the same.

Fix C — Thumb got a hoof, or the thumb is missing
------------------------------------------------------------
The thumb is a separate plain cream terry thumb with a rounded tip. It has NO hoof, cap, claw or nail. It angles out
about 25-30 degrees from the hand on the [RIGHT / LEFT] side, with white background between thumb and hand. Fix only
the thumb.

Fix D — Hoof is glossy, metallic or green
------------------------------------------------------------
The hoof finish is satin black paint with soft grey brush-streak highlights and a thin rim catching light-grey
highlights, like DETAIL_4. It is not glossy, metallic or iridescent and has no green sheen. Keep the shapes and
everything else the same.

Fix E — A fur cuff, sleeve or arm appeared
------------------------------------------------------------
```

```
Remove the [fur cuff / sleeve / arm / hand]. The glove ends at a plain, straight, open terry cuff edge at the wrist,
with nothing inside or around it. Keep everything else the same.

Fix F — Drew a pair
------------------------------------------------------------
Only one glove, please. Remove the second one and centre the remaining one. Keep everything else the same.

Fix G — Wrong angle (three-quarter, tilted or from above)
------------------------------------------------------------
This is a three-quarter / tilted / high view. I need a true [FRONT / SIDE] view: rotate the glove to exactly 90-degree
steps, stand it perfectly upright (hoof at the top, cuff at the bottom) and put the camera level with its middle,
looking straight across. Keep the glove itself identical.

Fix H — Terry looks like fur, fleece or knit
------------------------------------------------------------
The cream fabric is looped terry cloth, like a bath towel or a fluffy sock: small dense loops about 2-3 mm, like
DETAIL_3. It is not long fur, sherpa fleece or a knit. Keep the shapes and everything else the same.

Fix I — Proportions are off
------------------------------------------------------------
The proportions are off. The glove is about 2.8 times as tall as it is wide. The black hoof is the top ~36% of the
height and the cream terry is the lower ~64%. The thumb is short: about 4.5-5 cm on a 25 cm glove. Keep everything else
the same.

Fix J — Text, labels or a border appeared
------------------------------------------------------------
Remove all text, labels, numbers, arrows and borders. Pure white background only. Keep the glove exactly the same.

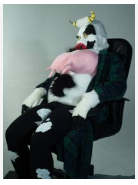
Detail images in this folder:
  - REF_C_worn_hoof_closed_from_above.png  (Closed hoof (for Fix A, B))
  - DETAIL_4_hoof_surface_brush_texture.jpg  (Hoof finish and rim (for Fix D))
  - DETAIL_3_terry_loop_texture.jpg  (Terry loops (for Fix H))
  - REF_A_left_glove_flat_back_of_hand.png  (Thumb shape and position (for Fix C))
  - DESIGN_KEY.png  (Design key (for any fix))

If two fixes in a row make it worse: start a new chat, run Step 1 again, attach your approved images, continue.
```

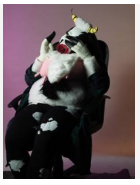
Appendix B. All figures (thumbnails; originals at full resolution in the zip)



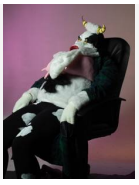
GLOVES-F001
left_flatlay_D800.jpg



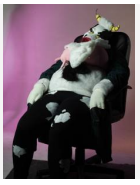
GLOVES-F002
costume_01.jpg



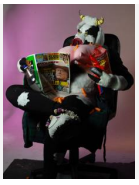
GLOVES-F003
costume_02.jpg



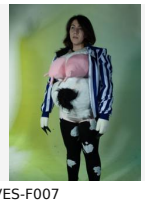
GLOVES-F004
costume_03.jpg



GLOVES-F005
costume_04.jpg



GLOVES-F006
costume_05.jpg



GLOVES-F007
costume_06.jpg



GLOVES-F008
costume_07.jpg



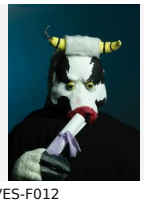
GLOVES-F009
costume_08.jpg



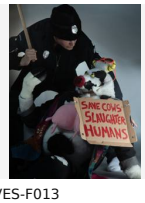
GLOVES-F010
costume_09.jpg



GLOVES-F011
costume_10.jpg



GLOVES-F012
costume_11.jpg



GLOVES-F013
costume_12.jpg



GLOVES-F014
costume_13.jpg



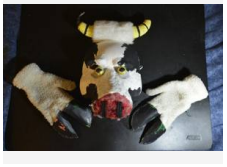
GLOVES-F015
right_alt_6D.jpg



GLOVES-F016
right_flatlay.jpg



GLOVES-F017
gloves_with_mask_photo1.jpg



GLOVES-F018
gloves_with_mask_photo2.jpg



GLOVES-F019
chatgpt_front_view.webp



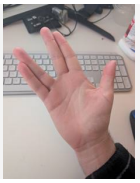
GLOVES-F020
chatgpt_front_view_resent.webp



GLOVES-F021
chatgpt_side_little_finger.webp



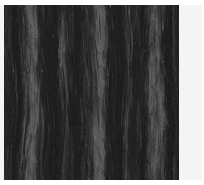
GLOVES-F022
chatgpt_side_thumb.webp



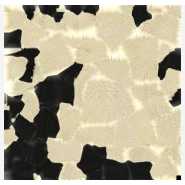
GLOVES-F023
user_hand_split_reference.jpg



GLOVES-F024
chatgpt_tile_terry.webp



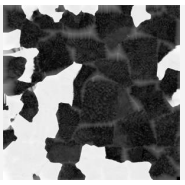
GLOVES-F025
chatgpt_tile_hoof_paint.webp



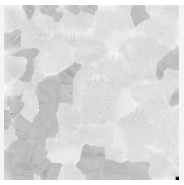
GLOVES-F026
tripo_v1_basecolor_8k.jpg



GLOVES-F027
tripo_v1_normal_4k.png



GLOVES-F028
tripo_v1_metallic_4k.png



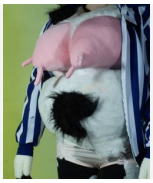
GLOVES-F029
tripo_v1_roughness_4k.png



GLOVES-F030
crop_a_worn_ribbon.jpg



GLOVES-F031
crop_b_raised_hand_V.jpg



GLOVES-F032
crop_c_jacket.jpg



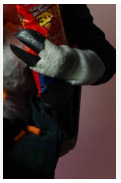
GLOVES-F033
crop_d_sheet.jpg



GLOVES-F034
crop_e_flat_toes_rim.jpg



GLOVES-F035
crop_g_floor.jpg



GLOVES-F036
crop_h_snack.jpg



GLOVES-F037
crop_i_terry_dirt.jpg



GLOVES-F038
crop_j_alt_image.jpg



GLOVES-F039
seg_test_isnet_failed.jpg



GLOVES-F040
seg_test_2.jpg



GLOVES-F041
seg_test_birefnet_ok.jpg



GLOVES-F042
refs_sheet_v1_notch.jpg



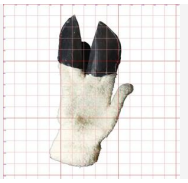
GLOVES-F043
refs_sheet_v2.jpg



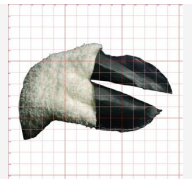
GLOVES-F044
REF_B_C_ribbon_notch.jpg



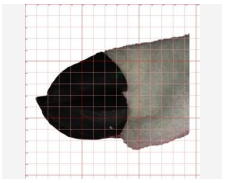
GLOVES-F045
REF_B_after_inpaint.jpg



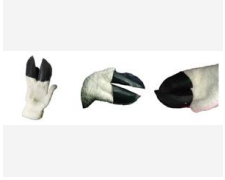
GLOVES-F046
grid_REF_A.jpg



GLOVES-F047
grid_REF_B.jpg



GLOVES-F048
grid_REF_C.jpg



GLOVES-F049
REF_ABC_final.jpg



GLOVES-F050
design_key_preview.jpg



GLOVES-F051
design_key.png



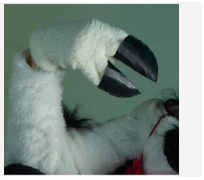
GLOVES-F052
REF_A_cutout.png



GLOVES-F053
REF_B_cutout.png



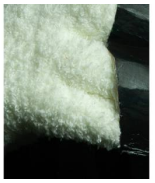
GLOVES-F054
REF_C_cutout.png



GLOVES-F055
DETAIL_1.jpg



GLOVES-F056
DETAIL_2.jpg



GLOVES-F057
DETAIL_3.jpg



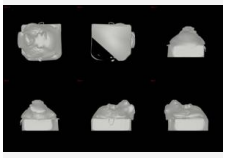
GLOVES-F058
DETAIL_4.jpg



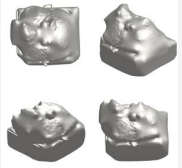
GLOVES-F059
DETAIL_5.jpg



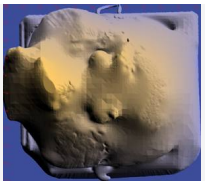
GLOVES-F060
guide_page_screenshot.png



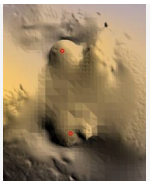
GLOVES-F061
mask_scan_views.png



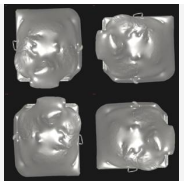
GLOVES-F062
mask_scan_top.png



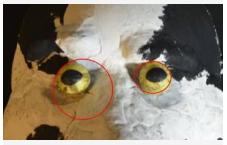
GLOVES-F063
mask_heightmap.png



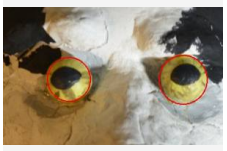
GLOVES-F064
mask_heightmap_eyes.png



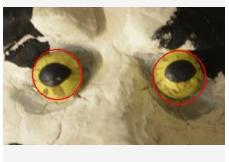
GLOVES-F065
mask_rotated.png



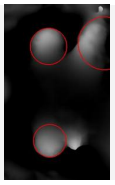
GLOVES-F066
photo2_eye_detection.png



GLOVES-F067
photo2_eye_circlefit.png



GLOVES-F068
photo1_eye_circlefit.png



GLOVES-F069
scan_eye_hough.png



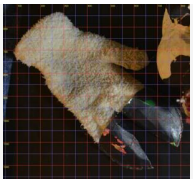
GLOVES-F070
glove_segmentation_photos12.jpg



GLOVES-F071
photo2_axes.jpg



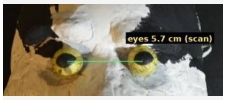
GLOVES-F072
photo2_grid_right.png



GLOVES-F073
photo2_grid_left.png



GLOVES-F074
scale_annotation_v1_bug.jpg



GLOVES-F075
eyeline_fixed.jpg



GLOVES-F076
scale_measured_photo2.jpg



GLOVES-F077
tripo_inputs_lineup.jpg



GLOVES-F078
tripo_input_front.png



GLOVES-F079
tripo_input_back.png



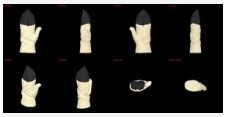
GLOVES-F080
tripo_input_left_thumb.png



GLOVES-F081
tripo_input_right_little.png



GLOVES-F082
tripo_inputs_lineup_sent.jpg



GLOVES-F083
tripo_v1_turnaround_textured.jpg



GLOVES-F084
tripo_v1_turnaround_clay.jpg



GLOVES-F085
tripo_v1_silhouette_check.jpg



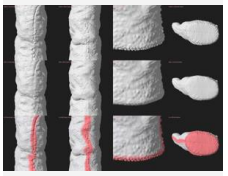
GLOVES-F086
tripo_v1_closeups_defects.jpg



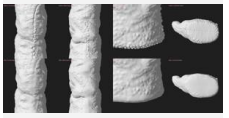
GLOVES-F087
highres_inspection_a.png



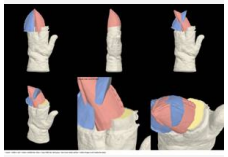
GLOVES-F088
highres_inspection_b.png



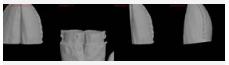
GLOVES-F089
seam_flatten_before_after_v1.jpg



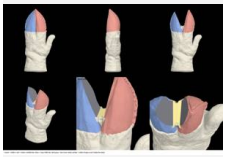
GLOVES-F090
seam_flatten_before_after.jpg



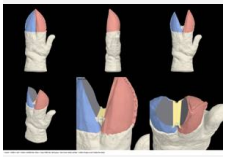
GLOVES-F091
split_r1_first_attempt.jpg



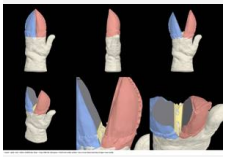
GLOVES-F092
split_r1_parts_separated.jpg



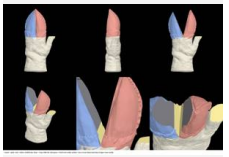
GLOVES-F093
split_r1_hoof_only_closed_spread.jpg



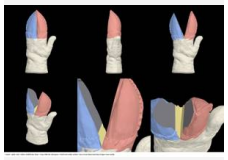
GLOVES-F094
split_r1_sent_file.jpg



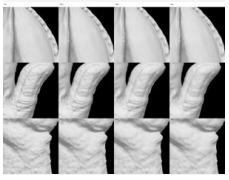
GLOVES-F095
split_r2_attempt1.jpg



GLOVES-F096
split_r2_attempt2.jpg



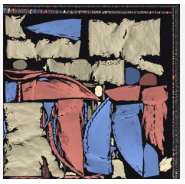
GLOVES-F097
split_r2_final_closed_spread.jpg



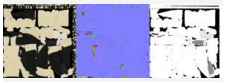
GLOVES-F098
decimation_150k_100k_64k.jpg



GLOVES-F099
uv_r1_smartproject_66.png



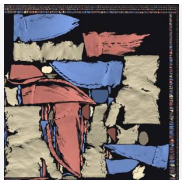
GLOVES-F100
uv_r2_smartproject_80.png



GLOVES-F101
bake_test_sheet_specks.jpg



GLOVES-F102
bake_test_preview.png



GLOVES-F103
uv_r3_smoothed_proxies.png



GLOVES-F104
bake_test_zoom_specks.png



GLOVES-F105
uv_r4_chart_merged.png



GLOVES-F106
bake_test2_clean.png



GLOVES-F107
bake_4k_preview.png



GLOVES-F108
pair_v1_render_a.jpg



GLOVES-F109
pair_v1_render_b.jpg



GLOVES-F110
spread_inner_wall_check.png



GLOVES-F111
pair_v1_render_final.png



GLOVES-F112
uv_mistake_specks_doc.jpg



GLOVES-F113
baked_4k_check_doc.jpg



GLOVES-F114
inner_walls_doc.jpg



GLOVES-F115
uv_atlas_v1v2.jpg



GLOVES-F116
v1_tex_color.jpg



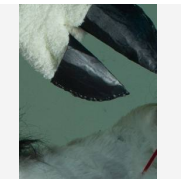
GLOVES-F117
v1_tex_normal.jpg



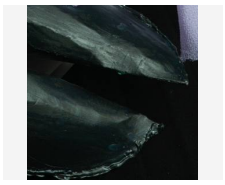
GLOVES-F118
v1_tex_roughness.jpg



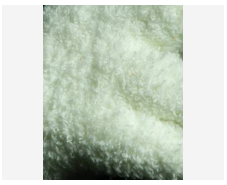
GLOVES-F119
swatch_paint_p1.jpg



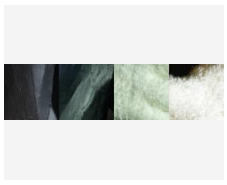
GLOVES-F120
swatch_paint_p3.jpg



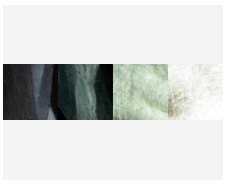
GLOVES-F121
swatch_paint_p2.jpg



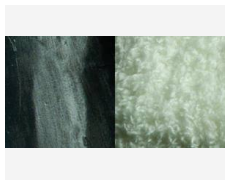
GLOVES-F122
swatch_terry_t1.jpg



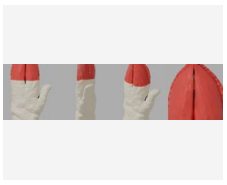
GLOVES-F123
swatch_sheet_v1.jpg



GLOVES-F124
swatch_sheet_paint.jpg



GLOVES-F125
swatch_sheet_terry.jpg



GLOVES-F126
region_map_v1.jpg



GLOVES-F127
region_map.png



GLOVES-F128
tilekit_terry_ref_a.jpg



GLOVES-F129
tilekit_terry_ref_b.jpg



GLOVES-F130
tilekit_paint_ref_a.jpg



GLOVES-F131
tilekit_paint_ref_b.jpg



GLOVES-F132
mask_paint_4k.png



GLOVES-F133
tiling_check_v2_first.jpg



GLOVES-F134
live_material_first_check.png



GLOVES-F135
pair_v2_render_a.jpg



GLOVES-F136
pair_v2_render_b.png



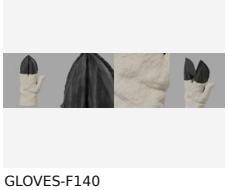
GLOVES-F137
live_vs_baked_v2.jpg



GLOVES-F138
moved_copy_test.png



GLOVES-F139
final_pair_v2.png



GLOVES-F140
final_closeups_v2.png



GLOVES-F141
v2_tile_paint_prepared.jpg



GLOVES-F142
v2_height_paint.jpg



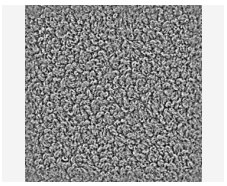
GLOVES-F143
v2_rough_paint.jpg



GLOVES-F144
v2_tiling_check.jpg



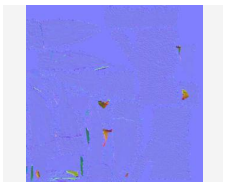
GLOVES-F145
tile_terry_prepared.jpg



GLOVES-F146
height_terry.jpg



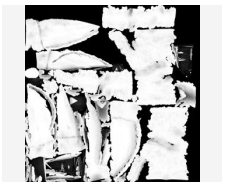
GLOVES-F147
v2_tex_color.jpg



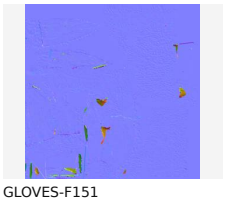
GLOVES-F148
v2_tex_normal.jpg



GLOVES-F149
v2_tex_roughness.jpg



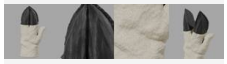
GLOVES-F150
v2_tex_ao.jpg



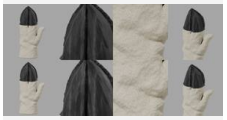
GLOVES-F151
v2_tex_detail_normal.jpg



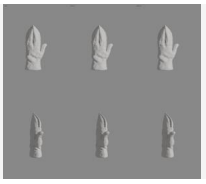
GLOVES-F152
v2_final_pair_jpg.jpg



GLOVES-F153
v2_final_closeups_jpg.jpg



GLOVES-F154
v2_check_live_baked_jpg.jpg



GLOVES-F155
smooth_perpart_clay_0_5_10.jpg



GLOVES-F156
smooth_perpart_clay_close.jpg



GLOVES-F157
smooth_perpart_clay_toepins.jpg



GLOVES-F158
smooth_perpart_web_pits.jpg



GLOVES-F159
smooth_unsplit_web.jpg



GLOVES-F160
before_after_v2_v3.jpg



GLOVES-F161
final_pair_v3.png



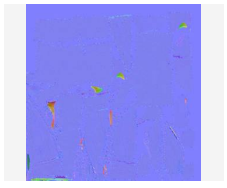
GLOVES-F162
inner_wall_v2_v3.jpg



GLOVES-F163
uv_atlas_v3.jpg



GLOVES-F164
v3_tex_color.jpg



GLOVES-F165
v3_tex_normal.jpg



GLOVES-F166
v3_tex_roughness.jpg



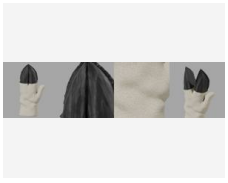
GLOVES-F167
v3_tex_ao.jpg



GLOVES-F168
v3_tex_detail_normal.jpg



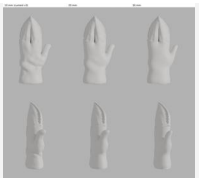
GLOVES-F169
v3_tex_mask_paint.png



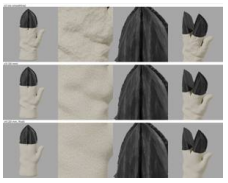
GLOVES-F170
v3_final_closeups_jpg.jpg



GLOVES-F171
v3_check_live_baked_jpg.jpg



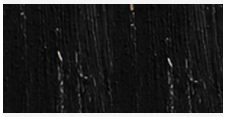
GLOVES-F172
smoothing_10_20_30_clay.jpg



GLOVES-F173
before_after_v4_specks_a.jpg



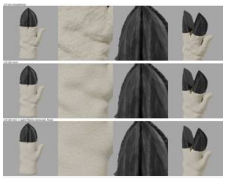
GLOVES-F174
final_pair_v4_specks_a.jpg



GLOVES-F175
paint_tile_lint_flecks.png



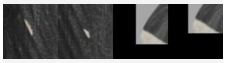
GLOVES-F176
paint_tile_fleck_fix.png



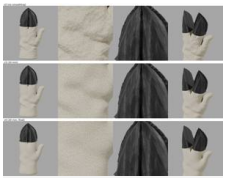
GLOVES-F177
before_after_v4_specks_b.jpg



GLOVES-F178
final_pair_v4_specks_b.jpg



GLOVES-F179
speck_render_crops.png



GLOVES-F180
before_after_v2_v3_v4_final.jpg



GLOVES-F181
final_pair_v4.png



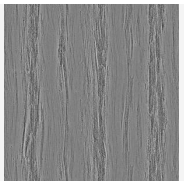
GLOVES-F182
live_vs_baked_v4.jpg



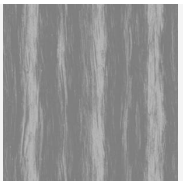
GLOVES-F183
uv_atlas_v4.jpg



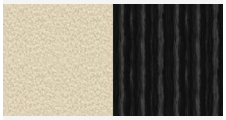
GLOVES-F184
v4_tile_paint_prepared.jpg



GLOVES-F185
v4_height_paint.jpg



GLOVES-F186
v4_rough_paint.jpg



GLOVES-F187
v4_tiling_check.jpg



GLOVES-F188
v4_tex_color.jpg



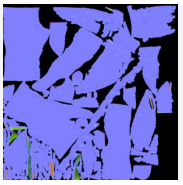
GLOVES-F189
v4_tex_normal.jpg



GLOVES-F190
v4_tex_roughness.jpg



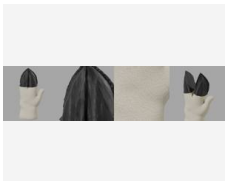
GLOVES-F191
v4_tex_ao.jpg



GLOVES-F192
v4_tex_detail_normal.jpg



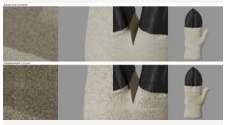
GLOVES-F193
v4_tex_mask_paint.png



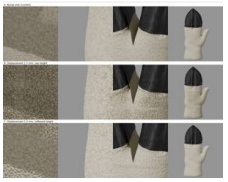
GLOVES-F194
v4_final_closeups_jpg.jpg



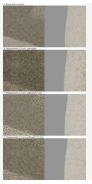
GLOVES-F195
displace_mask_preview.png



GLOVES-F196
bump_vs_displace_AB.jpg



GLOVES-F197
bump_vs_displace_ABC.jpg



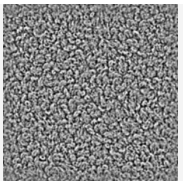
GLOVES-F198
bump_vs_displace_closeups_ABCD.jpg



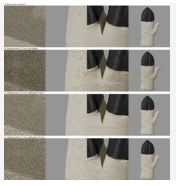
GLOVES-F199
final_pair_displacement.png



GLOVES-F200
displace_mask_4k.png



GLOVES-F201
height_terry_displace.png



GLOVES-F202
bump_vs_displace_ABCD_full.jpg



GLOVES-F203
integration_test_render.png